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REVIEW ARTICLE

A Systematic Review of Graph-Based Representation Techniques for Cyber-Attack Detection Across Application Domains

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Received: 27 January 2025 | **Revised:** 10 October 2025 | **Accepted:** 14 October 2025

Keywords: cyber-attack detection | cybersecurity | deep learning | feature extraction | graph representation | machine learning

ABSTRACT

Cyber-attack detection faces increasing challenges due to more advanced threats beyond traditional signature-based methods. Machine learning and deep learning have become essential in enhancing network security. Graph representation techniques capture structural and relational patterns across various data sources such as network traffic, system logs, and malware behaviors. Graph-based metrics and algorithms enable deeper analysis and more accurate classification of malicious activities. This study offers a systematic review of research articles published between 2019 and 2024 on graph-based representation methods for detecting cyber-attacks. Unlike previous reviews that classify studies by algorithm type or detection approach, this work groups them into four application domains: (i) general IDS/NIDS systems, (ii) botnet and DDoS detection, (iii) IoT-related threats, and (iv) advanced threats including Android malware, APTs, and encrypted traffic. Each domain is analyzed using graph modeling techniques, feature extraction methods (manual vs. automated), learning algorithms, and detection effectiveness. The review also introduces a five-stage framework outlining how graph-based methods can be integrated into cyber-attack detection: data acquisition, graph construction, feature extraction, detection/classification, and threat identification. This framework offers a structured view of methodological variety across different application areas. Results indicate that graph-based approaches provide a powerful alternative to traditional techniques, enabling richer structural modeling in cybersecurity. Finally, the study presents a future research roadmap advocating for interpretable, graph-driven solutions tailored to the evolving landscape of cyber threats.

1 | Introduction

As technology advances, the amount of data in cyberspace and the complexity of cyber-attacks targeting systems are increasing, making these threats more unpredictable. The rapid growth in processed data challenges security analysts to investigate attacks, analyze their characteristics, and assess their impact. Therefore, continuous monitoring and proactive investigation of malicious behavior are vital to protect information systems from cyber threats [1]. Cyber intrusion detection technologies are vital today, and various methods, such as intrusion detection systems, are commonly used in this field [2]. However, identifying cyber-attacks becomes more difficult as cyber threats and attack methods evolve. Traditional signature-based security solutions cannot effectively defend against advanced cyber-attacks or detect new attack method variants [3]. Signature-based detection systems are integrated into antivirus products and Intrusion Detection Systems (IDS) to identify network and system intrusions based on known attack patterns. However, these signature-based methods are limited because they do not adapt

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well to new attack types and can be bypassed by slight modifications to attack patterns during intrusions. Machine learning is increasingly used to overcome these limitations to improve intrusion and malware detection [4–8].

Artificial Intelligence (AI), encompassing machine learning and deep learning methods, has been extensively employed in cyber-attack detection in recent years and is expected to play a crucial role in future cybersecurity research. A substantial portion of the literature on cybersecurity focuses on studies utilizing machine learning and deep learning techniques. Moreover, there has been a significant increase in the proportion of studies employing deep learning techniques compared to those using traditional machine learning methods in recent years [9]. Furthermore, for deep learning and machine learning techniques to be effectively applied to datasets, those datasets must be appropriately formatted for compatibility with these models. Supervised machine learning algorithms, in particular, require a set of informative, discriminative, and independent features. In network prediction problems, it is essential to create a feature vector representation for nodes and edges [10]. Therefore, raw data that is not in a suitable format for machine learning techniques must undergo several preprocessing steps to extract meaningful information. Various methods are employed at this stage to analyze, classify, and group similar features within networks or datasets [11]. One of these methods involves transforming data into networked representations. Complex networks, formalizing the interconnected structure of various real-world systems such as biological systems, neural networks, social networks, computer networks, and so on [12, 13], can also enhance AI models' learning capabilities.

Recent interest has also emerged in data representation techniques that provide various methods for reconstructing data into higher-order representations. Learning methods relying on different data representations have been proposed, including examples such as Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and Generalized Discriminant Analysis (GDA) [14]. A concurrent and innovative approach to learning data representation is through graph representations. In recent years, increasing interest has been in using graph representations to model raw data before classification [15]. The primary motivation behind this approach is to extract representative subsequences that provide descriptive information and can aid in better understanding and managing complex networks [16]. Graph representation learning involves understanding and visualizing the graphical structure of a dataset and analyzing it using various machine learning algorithms. In this approach, mathematical models representing nodes and edges are developed based on the graph structure of the dataset. These models capture the properties of nodes and the relationships between edges. This representation method can subsequently be applied in machine learning tasks such as classification, clustering, and prediction [17]. Graph representation is widely used to model and analyze complex systems involving intricate relationships between system components. It offers numerous advantages such as flexibility, scalability, interpretability, and robustness. Graph-based models are flexible and can be adjusted to diverse structures and data types. Furthermore, graphs are exceptionally beneficial for capturing input data's inherent structure and relationships, presenting them concisely and interpretably. This capability enhances the efficiency of data processing and scalability [18]. Furthermore, graph-based models can offer richer insights into the relationships among system nodes, rendering them more interpretable than many traditional models. The simplicity and intuitive comprehensiveness of graph representation make it effective for modeling homogeneous and heterogeneous data from multiple sources, providing insights into communication networks or cyber-attacks [19, 20].

1.1 | Main Contributions

This study presents a comprehensive systematic review of graph-based, AI-assisted cyber-attack detection methods. The review focuses on research published between 2019 and 2024 and includes 41 primary studies selected through a rigorous multi-stage screening process. In the initial search phase, cybersecurity-related keywords such as "cyberattack," "intrusion detection," "botnet detection," "malware detection," and "DDoS detection" were combined with graph-related terms including "graph-based methods," "graph representation," "graph neural networks," "connectivity analysis," "graph mining," and "visibility graph." This search was conducted across major academic databases (IEEE Xplore, SpringerLink, ScienceDirect, Scopus, and Web of Science), yielding 211 articles. The overall workflow followed in this review is summarized in Figure 1, which outlines the six key stages from database search to the synthesis of domain-specific findings.

Following title and abstract screening, 114 studies were selected for initial review. These were further filtered based on established inclusion and exclusion criteria. The inclusion criteria specified that studies (i) be published between 2019 and 2024, (ii) be indexed in reputable academic databases, (iii) focus directly on graph-based cyber-attack detection, (iv) be written in English, and (v) have full-text availability. Exclusion criteria included lack of full text, non-English papers, poster abstracts, and publications outside the review scope. After full-text review, 41 eligible articles were retained for final analysis. The selected studies are grouped into four main application areas: (i) general IDS/NIDS systems, (ii) botnet and DDoS detection, (iii) IoT-related threat detection, and (iv) advanced threats like Android malware, APTs, and encrypted traffic analysis. Each category is examined based on the graph modeling approach (e.g., communication graphs, visibility graphs, and function call graphs), feature extraction method (manual vs. automated), machine or deep learning technique, dataset, and classification performance.

Rather than applying a rigid method-based taxonomy, this review adopts a domain-driven classification strategy and discusses graph-based techniques within each context. While some studies rely on handcrafted graph-theoretic features (e.g., degree and centrality), others employ end-to-end learning approaches using Graph Neural Networks (GNNs) such as GCN, GAT, or GraphSAGE. These differences are contextualized within each application area to highlight the diversity of methodological choices. Unlike existing reviews focusing on algorithmic categories or limited use cases, this study integrates classical and deep graph-based techniques into a unified review framework. Linking the graph representation type to specific cyber-attack

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scenarios provides a more practical and application-aware perspective on current research trends. Finally, several research gaps are identified, including the limited use of edge-level feature learning, insufficient exploration of graph-based methods for encrypted traffic and APT detection, and the lack of scalable and explainable models. These findings offer a roadmap for future work, underscoring the need for hybrid, robust, and interpretable graph-based solutions in next-generation cybersecurity systems.

As illustrated in Figure 1, the systematic review process in this study followed six stages: (i) identification of studies through database searches, (ii) title and abstract screening, (iii) full-text review of potentially relevant articles, (iv) application of inclusion and exclusion criteria, (v) classification of selected papers into predefined categories, and (vi) synthesis of findings across domains. This structured workflow ensured methodological rigor and transparency in article selection.

1.2 | Structure of the Article

This article is organized into six sections. Following the Introduction,

1. Section 2 provides foundational information on graph-based methods in cybersecurity, including graph representation strategies, traditional graph metrics, and deep learning approaches such as Graph Neural Networks (GNNs).
2. Section 3 describes the methodology of this systematic review, detailing the search strategy, inclusion and exclusion criteria, and the review process summarized in Figure 1.
3. Section 4 presents the main findings by analyzing selected studies across four application categories: general IDS/NIDS systems, botnet and DDoS detection, IoT-related threats, and advanced threats such as Android malware, APTs, and encrypted traffic. Each group is examined regarding graph construction methods, feature extraction strategies, model types, and performance results.
4. Section 5 offers a comparative discussion of the reviewed studies and highlights research gaps, limitations, and future directions.
5. Section 6 concludes the article by summarizing key insights and emphasizing the role of graph-based techniques in enhancing cyber-attack detection systems.

2 | Background

In this section, the study reviews the content focusing on machine learning, deep learning techniques, and graph representation techniques, emphasizing recent publications in the literature. It provides an overview and discussion of the studies that have been recently published on these topics, offering insights into their methodologies, findings, and contributions to the field of cyber intrusion detection.

2.1 | Machine Learning and Deep Learning Techniques

There are four different methods generally used in machine learning and deep learning: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. Labeled or classified data is used to train the model in supervised learning. This labeled data helps the model predict and make decisions. In unsupervised learning, there is no labeled data; the model is trained with input data, and no labeled dataset is used to support the learning process. Therefore, the data is grouped into clusters to create patterns and relationships. Semi-supervised learning is a hybrid learning model that includes both unsupervised and supervised learning methods. This method is used when only some of the labels of the input and output values are known. Reinforcement learning allows a system to learn from previous experience by observing the results of its interactions. This algorithm is a behavioral learning method that learns through trial and error, unlike supervised learning, which uses a sample dataset and guides users to the most accurate result based on the feedback it receives from analyzing the data [21].

In machine learning, the following algorithms are commonly used: Linear Regression, K-Nearest Neighbor Algorithm, Logistic Regression, Decision Tree, Random Forest, Support Vector Machines, and Naive Bayes Algorithm. Deep learning algorithms are generally composed of (but not limited to) Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Restricted Boltzmann

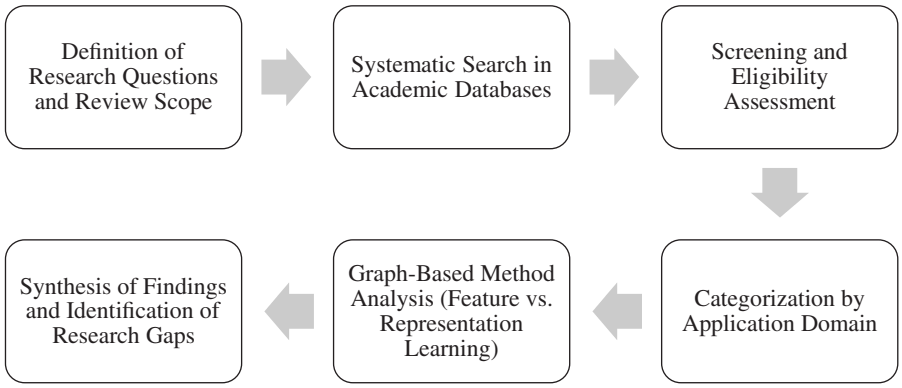


FIGURE 1 | Systematic review process employed.

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Machine (RBM), Deep Belief Network (DBN), Deep Autoencoders (Autoencoder), and so on. These algorithms serve various purposes across machine learning and deep learning domains, each suitable for different types of data and tasks [22]. In recent years, machine learning has emerged as a powerful tool for solving complex classification problems across various domains, including network security [23]. Indeed, machine learning and deep learning have numerous applications in cybersecurity, including antivirus software, email monitoring, network risk scoring, and phishing detection [24]. Thanks to artificial intelligence, antivirus software can detect previously unseen malware, model users' behavior based on login and logout patterns, and generate pre-attack alerts. Natural language processing techniques can scan suspicious emails, detect malicious code snippets running on servers, and identify various types of network attacks [25].

Below are examples of studies from the literature on cyber-attack detection using machine learning and deep learning (generally will be mentioned as AI in the rest of the article) techniques:

In the study by Shafiq et al. 2020, machine learning techniques were employed to prevent bot attacks on IoT networks. The study utilized the BoT-IoT dataset and identified 44 significant features for the machine learning algorithm. Five machine learning algorithms were employed to detect abnormal traffic [26]. Another study published by Shafiq et al. proposes CorrAUC as a feature selection method for detecting malicious bots in IoT networks using machine learning [27]. Pokhrel et al. utilized the K-Nearest Neighbor (KNN) and Naive Bayes (NB) algorithms on the Bot-IoT dataset to detect botnets in IoT networks. Their proposed model handles the security issue related to the threats from bots, where feature engineering and the synthetic minority oversampling technique (SMOTE) were combined with the mentioned machine learning algorithms (MLAs) [28]. Vinayakumar et al. used deep learning techniques for malware detection and classification with different datasets. The study proposes a new image processing technique with optimal parameters for machine learning algorithms and deep learning architectures. The experimental evaluations indicate that deep learning architectures perform better than traditional machine learning algorithms [29].

Popoola et al. designed a hybrid deep learning approach using Bidirectional Long Short-Term Memory (BLSTM) on the Bot-IoT dataset to detect bot attacks in IoT networks. Comprehensive experiments were conducted with the BoT-IoT dataset to validate the effectiveness of the proposed hybrid deep learning method. The experimental results indicate that it reduced the memory space required for storing large-scale network traffic data by 91.89% and outperformed existing feature dimensionality reduction methods by 18.92% to 27.03%. Furthermore, despite the significant reduction in feature size, the deep BLSTM model demonstrated robustness against underfitting and overfitting [30]. Vasan et al. presented a CNN approach for image-based malware classification, enabling the identification and classification of various types of malware. A novel classifier named IMCFN (Image-based Malware Classification using Fine-tuned Convolutional Neural Network Architecture) was proposed to detect malware variants and improve malware detection. Unlike existing solutions, a new method for multiclass classification problems is suggested, where the proposed method converts raw malware binaries into color images used by the CNN architecture. Experimental studies highlight that IMCFN outperformed other deep learning models, including other CNN models, with an accuracy of 98.82% on the Malimg dataset and over 97.35% on the IoT-Android mobile dataset [31].

An experimental study was presented by Meira et al. on various unsupervised learning algorithms used to detect unknown attacks using the NSL-KDD and ISCX datasets. The results of the experimental studies emphasized that the techniques used were capable of performing anomaly detection and can be integrated into future intrusion detection tools [32]. Alothman et al. introduced a new machine learning approach using the SMOTE technique to prevent botnet attacks, employing the Random Forest Algorithm in their proposed model. The study also tested various classifiers, and the results of the top three classifiers, namely J48, Random Forest (RF), and Multilayer Perceptron (MLP) networks, were reported. It was emphasized that the RF and J48 classifiers showed superiority compared to the MLP networks and other state-of-the-art solutions [33]. Thamilarasu et al. proposed an Intrusion Detection System (IDS) for IoT systems using a deep learning algorithm to identify malicious traffic, utilizing the NSL-KDD dataset. Experimental results indicated that the proposed intrusion detection system can effectively detect cyberattacks [34].

Kravchik et al. aimed to detect zero-day attacks on Industrial Control Systems using Convolutional Neural Networks (CNN) algorithms. The study proposed an anomaly detection method based on measuring the statistical deviation between the observed and predicted values. The dataset used contains 36 different cyberattacks, and it was stated that the proposed method successfully detected 31 attacks, thus providing an improvement over previous research based on this dataset [35]. Zhang et al. aimed to detect various web attacks (such as Brute Force, SQL injection, and XSS), infiltration, Heartbleed, Backdoor, and worms using Convolutional Neural Networks (CNN) deep learning techniques. They utilized the UNSW-NB15 and CICIDS2017 datasets for their study [36]. Asadi et al. proposed a botnet detection approach using LSTM, SVM, and Autoencoder to detect botnet attacks. This study implemented a method for detecting IoT botnet attacks using a combination of long short-term memory (LSTM), Autoencoder, and support vector machine (SVM) approaches. It was stated that the proposed approach achieved an improvement of 11.624% in accuracy, 11.629% in recall, and a reduction of 154.41 s in learning time for SVM [37].

Apostol et al. proposed an unsupervised deep learning approach for botnet detection using deep autoencoders. Also, a comprehensive search was conducted in multiple databases for primary studies published between 2018 and 2023. The results emphasize that machine learning and deep learning techniques demonstrate better performance in detecting IoT botnet attacks compared to traditional signature-based methods [38]. Ullah et al. developed a deep learning-based system using a Convolutional Neural Network (CNN) to detect anomalies in IoT networks. A convolutional neural network model was used to create a multiclass classification model, which was then implemented using 1D, 2D, and 3D convolutional neural networks. The proposed convolutional

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neural network model was validated using the BoT-IoT, IoT Network Intrusion, MQTT-IoT-IDS2020, and IoT-23 intrusion detection datasets. It is noted that the proposed models achieved high accuracy, precision, recall, and F1 score compared to existing deep learning implementations [39].

2.2 | Graph Representation Techniques

In network monitoring studies, a massive amount of data is generated, which poses challenges for analysis. Comprehensive approaches are necessary to filter, sample, and identify patterns in such data. Graph representation learning enables a deeper understanding of complex networked structures, and practical graph analysis can significantly enhance comprehension of these intricate structures [16]. Graph-based data representation facilitates the creation and analysis of graphs from big data, offering opportunities for network security and traffic analysis. This modeling approach allows large volumes of data to be analyzed and interpreted effectively [40]. Furthermore, graph-based deep learning is a rapidly evolving and promising research area. Recently, graph neural networks (GNNs) have achieved state-of-the-art performance in cybersecurity tasks such as network intrusion detection and anomaly detection [41, 42]. These developments enhance the eligibility of graph theory for analyzing and learning from complex systems. Graph theory has become an interdisciplinary field, from mathematical research to scientific applications, especially in network analysis, optimization, and computer science. In graph theory, graphs are defined as mathematical structures consisting of objects where specific pairs of objects are connected or related by edges. Graph theory is widely used to model and analyze relationships between objects or concepts [43]. Graphs are typically represented in the following format for a given graph G.

$$G = (V; E) \tag{1}$$

In Equation (1), V represents the set of vertices (nodes) representing objects, and E represents the edges (links) denoting connections between vertices. Nodes connected by an edge are called neighbor nodes. Graphs can take various forms: directed, undirected, weighted, and unweighted [13]. Visibility graphs, as an innovative conversion technique from the signal domain into the graph domain, represent the positions of a set of points in the Euclidean plane, where each point is tested for linear connectivity to determine if it can "see" other points [44]. Another definition describes a visibility graph as representing a set of distinct elements, each node depicts one element. An edge connects two nodes only if these nodes are mutually visible to each other [45].

The visibility graph proposed by Lacasa offers a simple and efficient computational framework that connects time series data with complex networks [46]. Visibility graphs have proven to be successful in several research areas, varying from image processing and physics to time series analysis, signal processing, and even social networks [45]. For any time series T, the amplitude values can be represented as a bar graph. When two data points are connected without intermediate obstacles blocking the line of sight, an edge is established between the corresponding nodes. This principle forms the basis of the visibility graph representation, as illustrated in the work of Lacasa et al. [47]. In addition, numerous visibility graph methods have been developed to transform time series or images into complex networks. Examples include the Natural Visibility Graph (NVG) and the Horizontal Visibility Graph (HVG), which can be implemented in different forms such as weighted/unweighted and directed/undirected [48]. Figure 2 displays visualizations of the Natural Visibility Graph and Horizontal Visibility Graph models [49]. Figure 2 demonstrates two common visibility graph transformations applied to time-series data: the Natural Visibility Graph (NVG) and the Horizontal Visibility Graph (HVG). In NVG, two data points are connected if a straight line drawn

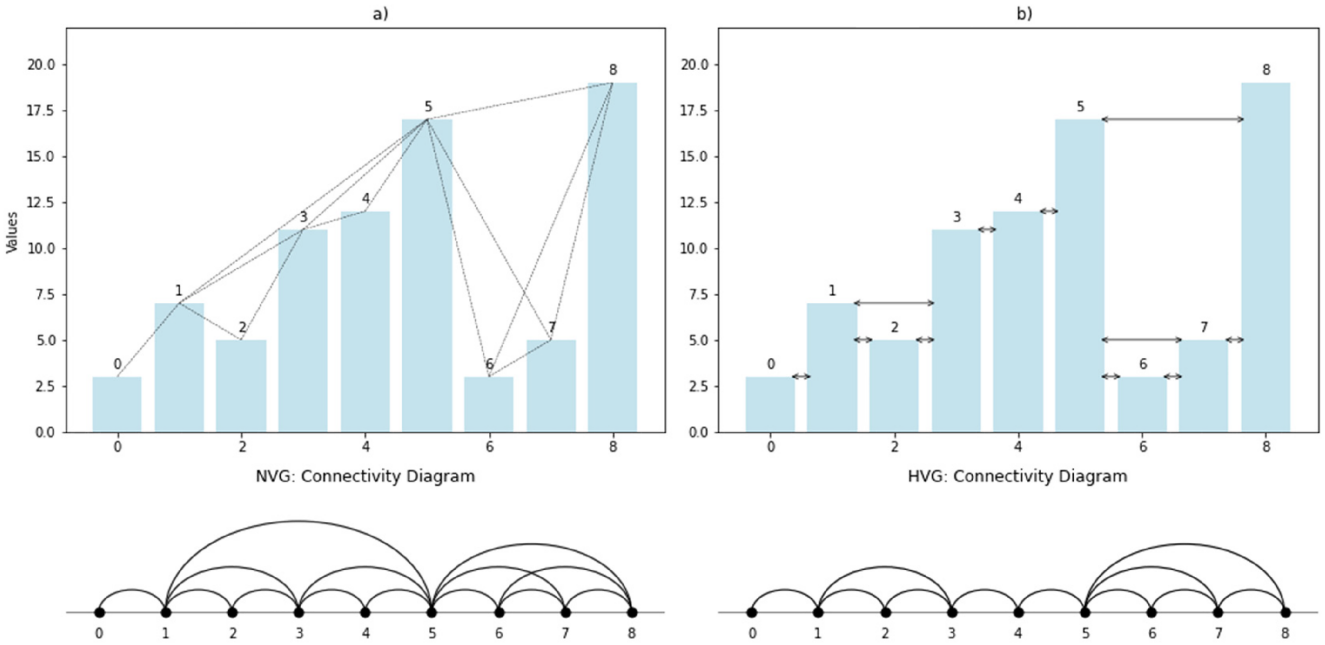


FIGURE 2 | (a) Natural visibility graph and (b) horizontal visibility graph.

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between them does not intersect any intermediate bar, thereby capturing global visibility patterns. In contrast, HVG applies a horizontal criterion, where nodes are connected only if no intermediate bar is higher, focusing on local connectivity. These transformations enable time-series signals to be represented as graphs, making them suitable for graph-theoretic analysis and subsequent machine learning tasks.

3 | Theoretical Framework: Mapping Graph-Based Methods to Cyber-Attack Detection Stages

This study introduces a conceptual framework aligned with five key stages of the cyber threat lifecycle to present a comprehensive view of the integration of graph-based methods within the cyber-attack detection pipeline. As illustrated in Figure 3, the process is structured into five functional layers. The first stage involves collecting raw data from diverse sources such as network traffic, system logs, sensor outputs, or binary malware files. Although often assumed in graph-based studies, this step is critical as it determines the quality and granularity of data available for graph modeling. For instance, many general IDS/NIDS studies rely on datasets such as CICIDS2018, UNSW-NB15, or BoT-IoT, which are derived from flow-level or packet-level network logs.

Figure 3 illustrates the proposed five-stage framework for integrating graph-based methods into the cyber-attack detection pipeline. The stages consist of: (i) data acquisition from network traffic, system logs, or malware samples, (ii) graph construction to model entities as nodes and their interactions as edges, (iii) feature extraction using graph-theoretic metrics or representation learning, (iv) classification and detection of anomalies through machine or deep learning algorithms, and (v) threat type identification, where detected anomalies are mapped to specific attack classes such as botnets, APTs, or DDoS. This conceptual pipeline highlights how graph-based approaches can systematically enhance each step of cyber-attack detection.

3.1 | Graph Construction

In this stage, raw or preprocessed data is transformed into a structured graph representation. The choice of graph type varies depending on the application domain:

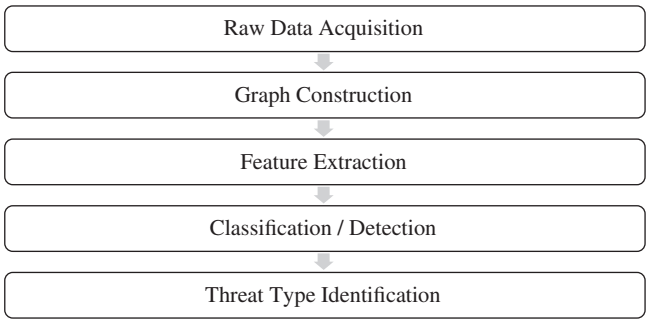


FIGURE 3 | Conceptual pipeline illustrating the integration of graph-based methods across the cyber-attack detection lifecycle.

- Function Call Graphs (FCG): Frequently used in Android malware analysis,
- Visibility Graphs (VG/NVG): Applied in time-series-based anomaly detection,
- Communication and Interaction Graphs: Common in botnet and DDoS analysis.

During this transformation, entities such as IP addresses, system functions, traffic packets, or behavioral events are modeled as nodes and edges to capture their relationships within the graph structure.

3.2 | Feature Extraction

Many traditional approaches extract structural graph-theoretic features such as node degree, clustering coefficient, betweenness centrality, eigenvector centrality, or graph density. These features are then fed into classical machine learning models such as SVM, Random Forest or KNN for classification. In particular, studies utilizing visibility graphs often compute these metrics from time-series signals. However, not all studies include this manual feature extraction step. More recent models based on Graph Neural Networks (GNNs) learn meaningful representations directly from the graph topology, eliminating the need for hand-engineered features.

3.3 | Classification and Detection

In the detection stage, classical and deep learning models are applied to distinguish between regular and malicious activity. While conventional classifiers such as KNN, MLP, or SVM are used in feature-driven studies, graph-based deep learning architectures—such as GCN, GraphSAGE, and GAT—have become increasingly popular for end-to-end learning from graph-structured data.

3.4 | Threat Type Identification

The final stage involves categorizing a detected anomaly into a specific threat class, such as botnet, APT, ransomware, encrypted traffic anomaly, or DDoS. This step is particularly emphasized in multi-class classification scenarios. For example, models such as STDeepGraph and BotMark detect attacks and distinguish between different threat types, thereby enabling higher-level situational awareness and response planning.

4 | Literature Review

The increasing complexity of cyber-attacks and the inadequacy of traditional security measures in addressing such threats have led to the growing prominence of artificial intelligence-based approaches in cybersecurity. Research on AI-assisted cyber-attack detection systems has gained significant momentum in recent years. Many of these studies apply classification algorithms to widely used benchmark datasets and report high accuracy rates. Model training and performance evaluation

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are typically based on statistical or behavioral features extracted from these datasets [50]. Indeed, Diro et al. proposed a new approach called deep learning in the field of cybersecurity for detecting attacks in the Internet of Things. It is stated that the proposed distributed attack detection system outperforms centralized detection systems by utilizing a deep learning model. The study presents a deep learning model reporting an accuracy rate between 95% and 99%. This high level of accuracy highlights the potential of deep learning techniques in enhancing security measures [51].

Zhao et al. present an approach for botnet detection using machine learning, highlighting high accuracy in detecting botnets with existing datasets. The study states that the Decision Tree classifier detected botnet activity in real-time with high accuracy. Experimental evaluations highlighted that an actual positive rate of over 90% and a false positive rate of under 5% were achieved [52]. Arwa et al. compare deep learning-oriented cybersecurity studies and report consistently high accuracy rates achieved with these methods. The study indicated that most of the proposed IDS in the cited papers use standard datasets such as KDD99 and its improved version, NSL-KDD. It is noted that 68% of the proposed solutions rely on benchmark datasets, while only 5% utilize real-time data obtained from simulated or real environments. Together, KDD99 and NSL-KDD account for more than 70% of the datasets the research community uses. Additionally, the article stated that Deep Belief Networks (DBN) performed better than Support Vector Machines (SVM), Naive Bayes, and Artificial Neural Networks (ANN), while the LSTM-RNN algorithm outperformed artificial neural networks (ANN), SVM, KNN, and Random Forest (RF) methods [53]. Gangavarapu et al. emphasize a model accuracy rate of 99% in their study on machine learning methods for spam and phishing mail filtering [54].

On the other hand, graph representation not only serves as a means of modeling network structures but also enables extracting meaningful features from these structures, thereby providing richer input to machine learning algorithms. Consequently, feature extraction processes conducted alongside graph-based representations have become crucial in enhancing detection performance.

Feature extraction from graph representations has emerged as a critical strategy to improve the effectiveness of cyber-attack detection systems. By leveraging network-theoretic measures such as degree, closeness centrality, betweenness, and clustering coefficient, researchers can derive informative features that reflect complex networks' topological and relational characteristics. The feature extraction process typically yields metrics such as the number of edges, link density, average closeness centrality, average degree, and clustering coefficient, which can be used as input to machine learning models. For instance, a degree is defined as the number of edges associated with a node and refers to the number of edges connected to a node [55], while betweenness is defined as the ratio of the shortest paths that pass through a node to the total number of shortest paths in the graph [56]. Proximity is related to the shortest path distances from a node to all other nodes [57] and eigenvector centrality is a measure of the influence of a node in a network and is defined as an extension of degree centrality [58].

Graph-based feature extraction is not limited to cyber-attack detection alone but has also yielded successful results in various other application areas. Below, several pioneering studies are presented in which feature extraction from graph representations has been effectively applied across different domains. These examples demonstrate the generalizability of the approach and set the stage for the following section, which focuses specifically on cybersecurity-related implementations.

In the study by Xinjie et al. 2021, vibration signals converted to the frequency domain were transformed into a visibility graph, and feature extraction was performed to represent the network derived from this graph. This study identified eight metrics: edge numbers, link density, average closeness centrality, average degree, graph entropy, average distribution weight, weight entropy, and clustering coefficient. After calculating these metrics, the training data were classified using the SVM algorithm [59]. In another study conducted by Wei in 2015, a new approach for feature extraction was presented, which involved transforming the dataset into complex networks to evaluate remote sensing image features. The dataset was transformed into a network, and a feature set was obtained by calculating some parameters out of this network. It was outlined that the applied method achieved better results compared to traditional methods [60]. In the study by Zhang et al. 2023, EEG signals were obtained and transformed into a visibility graph to detect early-stage Alzheimer's patients. The study identified 12 metrics to represent the graph. Specifically, six metrics were used: sequence similarity, average clustering coefficient, global efficiency, local efficiency, small-worldness, and graph index complexity. It was emphasized that the remaining six metrics, such as the size of the maximum clique, the cost of the traveling salesman problem (TSP), density, independence number, size of minimum cut, and vertex coloring number, are predominantly used in computer science and were not evaluated in the study [61].

In the study by Rashid et al. 2022, the Graph Convolutional Network (GCN) method was employed to detect COVID-19 patients. Computed Tomography (CT) images of the patients were transformed into a graph, and subsequently, the neighborhood matrix and degree matrix of the graph were constructed. Classification was then carried out using the GCN algorithm. The study also noted that GCNs, known for their superior classification performance, can be applied across various domains, including image classification studies [62]. In another study conducted by Rashid et al., a method is proposed for extracting feature information from computed tomography (CT) images using graph convolutional networks (GCNs) with different layer structures and kernel sizes to achieve a more accurate diagnosis of COVID-19. A U-Net model was applied to assist with segmentation and feature extraction. At the same time, GCNs were employed for classification and prediction to identify spatial connectivity patterns and benefit from these connections. Subsequently, the extracted feature information is processed by forming an adjacency matrix containing a graph structure and transferred to the GCN, the original image, and the largest kernel graph. The results of the study indicate that the proposed framework performs better in detecting lung diseases compared to recent deep learning architectures that are not based on graph representations [63]. In another study by Melo et al., focusing on feature extraction from audio signals, the audio signals were transformed into graphs, and metrics such as

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modularity, number of communities, average degree, and density were calculated for feature extraction. The study highlighted that this approach led to a higher success rate in the classification of audio signals [64].

The successful application of graph-based feature extraction methods across various domains has been demonstrated in the previous examples. However, the positioning of this approach—particularly in complex security scenarios such as cyber-attack detection—requires a dedicated examination. Recent studies have shown that graph-based representations can effectively model the structural relationships in multidimensional data such as network traffic, system calls, and software behaviors. Consequently, they significantly improve detection performance.

This section examines the literature on cyber-attack detection based on graph representations in detail. The presented review classifies the existing studies according to their methodological approaches and application domains under five main categories:

1. Review Studies on Cyber-Attack Detection Using Graph Representations.
2. Graph-Based Approaches for IDS/NIDS Systems.
3. Graph-Based Approaches for Botnet Detection.
4. Graph-Based Detection of IoT and DDoS-Based Attacks.
5. Semantic Behavior-Based Attacks: Detection of Android Malware and APTs.

4.1 | Reviews on Cyber-Attack Detection With Graph Representation Techniques

In recent years, graph representation techniques have garnered increasing attention within the field of cyber-attack detection, leading to a proliferation of review articles that map the evolving methodological landscape. These surveys offer valuable overviews of how graph mining, graph-based deep learning models, and graph-driven anomaly detection techniques have been integrated into various cybersecurity applications. By synthesizing insights across different approaches and threat domains, they contribute to a better understanding of the role and potential of graph-based methods in security research. This section summarizes the key review studies in this area, focusing on their methodological perspectives, application scopes, and highlighted trends.

Pazho et al. systematically review graph-based deep learning approaches for anomaly detection in distributed systems. The study emphasizes the importance of modeling complex interactions using graph structures in modern distributed environments, such as microservice architectures, cloud computing, and container-based infrastructures. Techniques employed for anomaly detection are presented and compared, including GNNs, autoencoders, Transformer-based architectures, and graph embedding methods. Additionally, the study focuses on topics such as modeling monitoring data as graphs, learning objectives at the node and subgraph levels, and the detection of time series-based anomalies [18].

In the study conducted by Lagraa et al., the use of graph-based methods in network security is comprehensively examined. The study reviews recent research in areas such as graph representation of network traffic, botnet detection, intrusion detection systems (IDS), and the analysis of suspicious authentication behaviors. Methodologically, it encompasses both classical graph mining approaches and graph-based deep learning techniques. Furthermore, the study provides a detailed comparison of the types of graphs used, types of features extracted, data sources, and categories of security threats. In its conclusion, the study also addresses the main challenges faced by graph-based security approaches (e.g., scalability, interpretability, and data quality) and outlines potential directions for future research [40].

In the study conducted by Yan et al., both classical graph mining techniques and deep learning-based graph approaches (e.g., Graph Neural Networks—GNNs) used in cybersecurity are comprehensively examined. The research details the main cybersecurity tasks in which graph-based methods are applied and focuses on critical threat scenarios such as anomaly detection, authentication analysis, APT attack monitoring, botnet detection, malware analysis, financial fraud, and fake account identification. The study adopts a broader perspective by including not only attack detection but also graph mining applications targeting various cybersecurity tasks such as fraud detection, identity spoofing, and authentication [65]. In the study by Jiang et al., graph-based deep learning methods in communication networks are comprehensively addressed. This review examines how Graph Neural Network (GNN) architectures are applied in wired and wireless communication systems (e.g., data centers, optical networks, IoT, and SDN). Common GNN variants such as GCN, GAT, GraphSAGE, and GIN are explained, and their applications in areas such as network performance management, routing, resource allocation, and service placement are detailed. Moreover, the study highlights fundamental challenges, including scalability, real-time processing, and explainability, and offers suggestions for future research directions [66].

In the study conducted by Bilot et al., Graph Neural Network (GNN)-based approaches are systematically examined, particularly in the context of network-based intrusion detection. The study compares existing GNN architectures (e.g., GCN, GAT, and GraphSAGE) in supervised and unsupervised learning, and evaluates their applications in IDS/NIDS systems. Furthermore, the article summarizes the main benchmark datasets used (e.g., NSL-KDD, CICIDS2017, and TON_IoT) and analyzes the performance achieved by GNNs on these datasets. The study also discusses the advantages and challenges of GNN models in terms of data preprocessing, feature selection, and model complexity [67]. The study by Yang et al. provides a comprehensive review of Advanced Persistent Threat (APT) attack detection methods based on provenance graphs. Provenance graphs, which record detailed causal and temporal dependencies between system entities and events, offer a powerful representation to model complex multi-stage APT attacks. The review categorizes existing approaches according to graph construction techniques, feature extraction methods, and detection algorithms, including machine learning, deep learning, and graph neural networks. Various studies employing structural, temporal, and semantic graph features are analyzed. Performance metrics commonly reported in the literature include accuracy, precision, recall,

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F1-score, and detection latency. Reported accuracy rates for APT detection methods using provenance graphs typically range between 85% and 98%, depending on the dataset, attack scenarios, and algorithms employed. Graph-based methods outperform traditional signature- or rule-based systems by better capturing the subtle correlations and multi-step attack behaviors inherent in APTs. Challenges identified include scalability to large datasets, real-time detection capabilities, and effective handling of imbalanced attack classes. The article highlights future research directions focusing on dynamic graph analysis, explainability of detection results, and integration with broader cybersecurity frameworks [68].

In the survey by Bilot et al., a comprehensive and methodologically structured review is presented on the application of Graph Neural Networks (GNNs) for network-based and host-based intrusion detection. The study emphasizes that traditional ML/DL models relying on flat data representations fall short in capturing the relational and structural behaviors of cyber-attacks—particularly for complex and stealthy threats such as Advanced Persistent Threats (APTs) and botnets. In contrast, graph-structured data—such as flow graphs, packet graphs, and authentication graphs—enable a more semantic and robust representation of interactions between system entities. The review systematically categorizes graph types (e.g., heterogeneous, spatio-temporal, and dynamic), feature extraction approaches (e.g., meta-paths and random walks), and GNN architectures, including GCN, GAT, GraphSAGE, HAN, and GIN. Detection tasks are mapped to node-level, edge-level, and graph-level classification depending on the attack target (e.g., malicious host, flow, or session). Reported results across datasets such as CIC-IDS2017, CTU-13, UNSW-NB15, and ToN-IoT demonstrate superior detection accuracy (often above 90%) and robustness against adversarial attacks compared to conventional methods. The article also evaluates self-supervised and semi-supervised learning paradigms and highlights GNNs' ability to generalize without domain-specific feature engineering. Key challenges include scalability to large graphs, real-time graph stream processing, and interpretability of predictions. Future directions involve integrating temporal dynamics, improving explainability, and leveraging heterogeneous graph attention mechanisms for fine-grained intrusion analysis [69].

A list of review studies on graph-based approaches is presented in Table 1, which includes information on the methods used, application domains, and aspects that differentiate these studies from our work. The table provides a comparative overview of each review's focus on graph techniques, targeted cybersecurity tasks, and classification strategies, thereby offering a clearer perspective on existing gaps and emerging trends in the literature. While most existing reviews focus on specific algorithmic techniques or narrow application domains, this study introduces a holistic classification that considers both threat types (e.g., IDS/NIDS, botnet, IoT/DDoS, Android malware, and APT attacks). Another key contribution of this review is its emphasis on the practical significance of graph representations. Rather than grouping studies solely by the techniques used, it also categorizes them based on the data type analyzed (e.g., network

TABLE 1 | A list of review articles in AI-assisted cyber-attack detection with graph-based approaches.

References	Overview	Year	Method	Detection focus	Application level	Key difference
[18]	Graph deep learning for anomaly detection in distributed systems	2022	Autoencoders, Graph Embedding, Transformers	Anomaly detection in distributed systems	System level (Distributed)	Deep Learning Only
[40]	Overview of graph-based methods for network security	2023	Graph mining, graph learning	IDS, botnet, suspicious authentication	Network level	General Scope
[65]	Survey on classical and deep graph mining techniques applied to cybersecurity	2023	Graph statistical features, graph embeddings	Malware, intrusion, botnet, fraud, fake accounts, etc.	Network level, behavioral	Broader Scope
[66]	Graph-based deep learning in communication networks	2022	GCN, GAT, GraphSAGE, GIN	Network performance and security	Network level	Communication Inf. Focus
[67]	GBAD methods for fraud detection	2020	Community-based, structural, probabilistic GBAD	Fraud detection	Transactional	Financial-oriented
[68]	Provenance-based graph classification for APT detection	2022	Provenance graph construction, ML/DL classifiers	APT detection	System level	APT-specific
[69]	GNN-based techniques for intrusion detection	2023	GNN architectures (GCN, GAT, etc.)	IDS/NIDS	Network level	GNN-Centric

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traffic, system calls, and behavioral traces) and the threat models addressed. This allows for a more precise assessment of how graph-based methods perform under varying operational conditions. Finally, the review also identifies existing gaps in the literature. For example, the limited exploration of intersections between specific graph techniques (e.g., VG/NVG) and advanced threat types (e.g., APTs and Android malware) points to concrete directions for future research. In this respect, the review synthesizes current work and offers a practical roadmap for researchers and cybersecurity professionals. It provides a more holistic perspective on the evolving role of graph-based methods in cybersecurity. It serves as a guide for applying graph intelligence in real-world cyber defense systems.

Having demonstrated the effectiveness of graph-based feature extraction across various domains, the focus now shifts to its contributions and applications within the cybersecurity context. Due to their ability to model network dynamics, behavioral patterns, and system-level interactions, graph-based representations have become particularly well-suited for a wide range of cyber-attack detection scenarios. The following sections systematically examine how graph-based approaches are employed across different cybersecurity domains. This structured classification highlights the diversity of application areas and the evolving methodological trends in the field.

4.2 | Graph-Based Approaches for IDS/NIDS Systems

Early detection of network-based attacks is a critical requirement in cybersecurity. Traditional Intrusion Detection Systems (IDS/NIDS) typically rely on signature-based or statistical approaches, often falling short when faced with zero-day attacks and complex anomaly patterns. To overcome these limitations, graph-based representation methods have emerged as promising alternatives by comprehensively modeling the structural and temporal relationships within network traffic. In this context, graph theory and Graph Neural Networks (GNNs) have become increasingly prevalent for more accurate and timely detection of network-based threats. In the reviewed studies, network traffic is generally represented through graph structures, where nodes correspond to entities such as devices or sessions, and edges represent communication links between them. Feature extraction is conducted through two main strategies: (i) direct representation learning on the graph structure using methods such as GNNs or autoencoders, or (ii) extraction of graph-theoretic metrics—such as degree, betweenness centrality, and clustering coefficient—which are then used as input to traditional machine learning classifiers. Among the most commonly used datasets in these studies are CICIDS2017, UNSW-NB15, and ISOT-CID. Numerous studies have explored integrating graph-based representations within IDS/NIDS frameworks. Table 2 summarizes original research articles that apply AI-driven graph techniques to intrusion detection systems.

Alwasel et al. conducted a comprehensive experimental study that included data preprocessing, visualization, feature analysis, and comparison of machine learning models to assess the effectiveness of graph representation in enhancing classification accuracy. The study involved transforming network traffic data into a graph-based representation, where nodes represented devices and edges represented communication instances. These graph features were then used as input for machine learning models. The study followed several steps, including data processing, graph-based data representation using the neighborhood matrix method, feature extraction, classification with machine learning algorithms, and performance evaluation. Various graph construction techniques, such as adjacency matrices and distance-based methods, were explored to determine the most suitable representation for the dataset. After creating graph-based representations, feature extraction and dimensionality reduction using Principal Component Analysis (PCA) were performed to enhance computational efficiency. The ISOT Cloud Intrusion Detection Dataset (ISOT-CID) was utilized in the study. The study concluded that incorporating graphs into network data analysis significantly enhances machine learning models' performance. Specifically, the study reported that while accuracy ranged from 90% to 96% without using graph representation techniques, it increased from 97% to 99% when using graph representation techniques. Ultimately, the study goes beyond traditional feature extraction by structurally modeling attack flows through graph similarities and integrating this information with deep learning. This approach provides high accuracy and interpretability, particularly in detecting complex and low-frequency attacks [23].

The study conducted by Liyan et al. proposes two novel graph-based solutions for intrusion detection: a modified E-GraphSAGE and an edge-based residual graph attention network (E-ResGAT). These methods extend existing GNN models by integrating residual learning to address the significant class imbalance in cybersecurity datasets, enhancing minority class detection performance. Using network flow data mapped into bipartite and line graph structures, the models effectively capture both node and edge features. Experimental evaluations on four recent intrusion detection datasets demonstrate that the proposed models outperform their original counterparts, especially in classifying minority attack classes, while maintaining strong overall accuracy. The study also highlights the advantages of attention mechanisms in improving model robustness. Future work aims to optimize graph neighborhood construction and tackle more imbalanced datasets through synthetic data augmentation [41].

Zhu et al. proposed a Graph Neural Network (GNN)-based Network Intrusion Detection System (NIDS) to explore the structural information of networks and the relationships between traffic patterns. In their approach, traffic data was represented using a graph structure to capture topological information. Given that traffic with similar characteristics or originating from the same source/destination may produce similar representations in GNNs, the study introduced a two-part Graph-Based Behavior Analysis Intrusion Detection System (GBA-IDS). In the first part of GBA-IDS, the Line-GraphSAGE model was employed to learn the network's topological structure. This model utilized aggregation operations to gather information from neighboring edges, generating edge representations within the graph. The topological structure of each node's neighborhood and the distribution of node features were obtained by extracting feature information from a node's neighbors. Line-GraphSAGE was trained using traffic flow features and the graph as inputs, generating

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TABLE 2 | A list of research studies employing graph-based AI methods for (IDS) NIDS systems.

References	Overview	Classification methods	Dataset	Year	Domain	Threat type
[23]	Network traffic data was converted into a graph representation, followed by the creation of its neighborhood matrix, and subsequently, feature extraction using Principal Component Analysis (PCA) was performed	RL, SVM ve K-means + PCA	ISOT Cloud Intrusion Detection Dataset	2023	IDS/NIDS	Intrusion detection
[41]	Two GNN-based models, *Modified E-GraphSAGE* and *E-ResGAT*, are proposed to address class imbalance in intrusion detection, using bipartite and line graph structures for learning from both node and edge features	Modified E-GraphSAGE, E-ResGAT	IDS Datasets	2023	IDS/NIDS	Intrusion detection
[70]	A Graph Neural Network-based Network Intrusion Detection System (NIDS) is designed, which includes a Graph-based Behavior Analyzing IDS (GBA-IDS) with two parts: Line-GraphSAGE and Discriminator	Line-GraphSAGE + Discriminator	TON IoT, BoT-IoT, UNSW-NB15	2022	IDS/NIDS	Intrusion detection
[71]	A hybrid deep neural network, combining CNN and LSTM, named STDeepGraph, is proposed	CNN + LSTM (STDeepGraph)	UNSW-NB15, CIC-IDS-2017	2019	Flow-based detection	IDS/NIDS
[72]	A detection method based on graph kernels and CNNs is proposed	Graph Kernels + CNN	UNSW-NB15, CIC-IDS-2017	2019	IDS/NIDS	Intrusion detection
[73]	A new autoencoder-based graph representation method called Security Objects' Graphs is proposed	Auto-encoder	CICIDS2017	2020	IDS/NIDS	Anomaly detection
[74]	A three-step methodology was employed: constructing a graph from the data, applying data preprocessing steps to the graph data (using R and SM-hybrid methods), and performing classification with a Neural Network and Graph Convolutional Network (GCN)	CNN + GRU	UNSW-NB15	2022	IDS/NIDS	Intrusion detection
[75]	A new method, Deep Graph Feature Learning (DeepGFL), is proposed for modeling network graphs	DeepGFL (CNN)	CIC-IDS-2017	2018	IDS/NIDS	Intrusion detection
[76]	Logs are transformed into a graph, and 21 features are extracted using One-Hot-Encoding	GNN + PCA + Clustering	Electra, HAI	2023	IDS/NIDS	Critical infrastructure attacks
[77]	A GNN-based anomaly detection model named *Anomal-E* is proposed, which learns embeddings from edge-attributed graphs without manual feature extraction, achieving high F1 scores on benchmark intrusion datasets.	GNN	NF-UNSW-NB15, NF-CSE-CIC-IDS2018	2022	IDS/NIDS	Intrusion detection
[78]	Unsupervised GNN-based autoencoder model that learns representations from IP-level network graphs to detect anomalies without relying on labeled data	GNN	NSL-KDD	2024	IDS/NIDS	Intrusion detection

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edge representations matching the original feature size. The second part involves a discriminator, which takes the edge representations produced by Line-GraphSAGE as input and outputs predictions. The Discriminator aimed to refine the edge representations generated by Line-GraphSAGE to improve overall model performance. Three datasets—TON IoT, BoT-IoT, and UNSW-NB15—were utilized in the study. The results demonstrate that the model improved significantly in binary and multiclass classification tasks, with accuracy rates ranging from 97% to 100%. In conclusion, the study achieves a more comprehensive behavioral representation by modeling the content features of network flows and the inter-node relationships. Flow-level representation learning was performed using Line-GraphSAGE, and a discriminator component was incorporated to focus on misclassifications, thereby improving overall accuracy [70].

Yao et al. focused on designing a hybrid deep neural network named STDeepGraph, which combines Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) architectures to learn high-dimensional representations from network traffic data. In this study, a novel approach is presented for flow-based attack detection by integrating structural similarities learned through graph kernels with a deep learning architecture, thereby enabling the use of both structural and contextual information. In their study, a graph was constructed from network flow data, and a kernel matrix was derived from it. This kernel matrix serves as input to both the CNN and LSTM algorithms. The study involves analyzing sequences of network flows, constructing temporal communication graphs, and calculating graph kernel matrices. These kernel matrices capture the relationships between graphs, and graph vectors are computed based on these matrices. The UNSW-NB15 and CIC-IDS-2017 datasets were used, with variations that include datasets with and without feature extraction. The proposed method achieved 98.6% and 99.4% accuracy rates on the respective datasets. Experimental findings indicate that the approach effectively extracts discriminative representations, improving accuracy in supervised classification tasks. The overall accuracy of the method was shown to improve by approximately 10% to 15% in experiments [71].

Su et al. focused on a graph kernel deep learning architecture. They created a graph kernel matrix from network flow metadata, which was then used as input for the CNN algorithm. Structural similarity matrices were generated using the shortest path kernel to measure similarities between graphs for feature extraction. The study utilized two datasets, UNSW-NB15 and CIC-IDS-2017. The proposed method achieved higher accuracy, precision, and F1 scores than baseline CNN and LSTM models on both datasets. For instance, the F1 score reached 98.7% on CIC-IDS-2017 and 99.3% on UNSW-NB15. These results demonstrate that integrating graph kernels provides a significant performance improvement. The evaluation results validate the accuracy of the proposed framework and the successful integration of deep learning and graph kernel methods in cybersecurity. In this study, promising results are achieved by combining structural similarities extracted through graph kernels with LSTM- and CNN-based deep learning architectures [72].

Leichtnam et al. proposed a novel auto-encoder-based graph representation method called "Sec2graph." Their study constructed a graph using network metadata and introduced an unsupervised learning approach based on auto-encoders for anomaly detection within the graph structure. The graph was converted into a matrix representation, and then anomaly detection was performed. Raw log files were initially processed into network and application data using the Zeek IDS tool. A graph representation was built, but no additional feature extraction was performed; the graph was directly used as input to the model. The CICIDS2017 dataset was used in their experiments. The proposed method achieved a high actual positive rate (97.6%) by enabling system logs to be represented with more meaningful structures through graph transformations [73].

Zola et al. presented a temporal graph-based approach for detecting cyber-attacks in network traffic data. The proposed method consists of three main stages: (i) generating temporal graphs from multivariate time-series, (ii) applying data preprocessing techniques (e.g., R and SM-hybrid) to address class imbalance, and (iii) performing classification using a hybrid deep learning model combining Graph Convolutional Networks (GCNs) and Gated Recurrent Units (GRUs). Rather than relying on manual extraction of graph-theoretic features (e.g., degree and centrality), the model automatically employs GCNs to learn node-level representations from graph structure. Experimental results show that the proposed method achieves an F1-score exceeding 95%, outperforming traditional LSTM and CNN models in accuracy and false positive rate. The study highlights that temporal dissection parameters strongly influence detection performance, and that data-level balancing improves learning effectiveness. Overall, this work introduces a novel architecture that captures both spatial and temporal aspects of attack patterns through graph-based deep learning, without requiring explicit feature engineering [74].

Yao et al. introduced a novel method called Deep Graph Feature Learning (DeepGFL) for modeling network graphs. This approach generalizes high-order features from raw features extracted from generated graphs and applies them to network intrusion detection. The study begins with feature extraction from raw network traffic (PCAP files). Using the CICFlowMeter, 80 features were initially extracted. Subsequently, the eight most critical features were selected: flow duration, total forward packets, total backward packets, total length of forward packets, total length of backward packets, flow bytes per second, flow packets per second, and down/up ratio. Using the CIC-IDS-2017 dataset, the study constructs graphs from the extracted data and performs classification in the final stage. Experimental results demonstrate that DeepGFL achieves superior effectiveness, accuracy, and space efficiency in network intrusion detection. This study presents a novel deep learning framework that relies on raw features and incorporates representation learning based on high-level graph relationships by constructing graph representations from network traffic. DeepGFL successfully modeled complex relationships in network traffic through graph-derived features and achieved high detection accuracy in identifying attacks. Furthermore, the study provides a comparative analysis of traditional and graph-based feature selection strategies. In this respect, DeepGFL introduces a hybrid approach that integrates classical machine learning with graph-based learning [75].

Desnitsky et al. proposed an approach for detecting attacks on critical infrastructures using graph-based modeling and machine

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learning. The proposed approach was tested on two datasets, Electra and HAI. For the Electra dataset, numerical features were normalized through scaling, while categorical features were processed using one-hot encoding, resulting in 21 features. Additionally, a hierarchical clustering method was employed to group similar real-world situations. A vital feature set regarding the HAI dataset was identified using Principal Component Analysis (PCA). Initially, the optimal number of principal components needed to explain at least 90% of the variance in the data was determined. Subsequently, the most significant features for each principal component were obtained by calculating the covariance matrix, identifying 17 features. This study presents an original framework based on modeling state changes in critical infrastructures using graph structures. By leveraging a GNN architecture, learning is performed directly on the graph, enabling effective classification of attack patterns within complex system dynamics. It significantly contributes to threat detection based on graph representations, particularly in specialized domains such as SCADA systems [76].

In 2022, Caville et al. introduced Anomal-E, a Graph Neural Network (GNN) approach designed for intrusion and anomaly detection, leveraging the topological structure of graphs. The study performed no explicit feature extraction; instead, automatic embeddings were learned directly from the graph structure and edge attributes. The edge features were obtained from network flows. The study utilized the NF-UNSW-NB15 and NF-CSE-CIC-IDS2018 datasets. They reported a substantial performance improvement when employing Anomal-E compared to raw features and other conventional algorithms. The proposed method has demonstrated significantly superior performance across all metrics. For example, an F1 score of 88.45% was achieved on the NF-UNSW-NB15-v2 dataset. Additionally, it outperformed similar GNN-based methods such as GraphSAGE and DGI. In conclusion, it is the first GNN-based NIDS approach focusing on edge features and has made substantial advancements in both model performance and generalizability [77]. In the study conducted by Sathishkumar et al., the goal was to detect anomalies in network security using an unsupervised Graph Neural Network (GNN) approach. A graph structure was constructed based on IP-level connection relationships, and direct representation learning was performed using an autoencoder architecture. By leveraging structural information at both the node and edge levels, anomalies in network traffic were successfully distinguished without the need for any label information. Experiments conducted on the NSL-KDD dataset reported an F1 score of 86% for anomaly detection, with a ROC-AUC value of 93%. In this regard, the study contributes to label-independent cyber-attack detection through unsupervised graph learning [78].

Graph-based methods applied to IDS/NIDS systems offer richer and more effective representations for cyber-attack detection by incorporating structural relationships within network graphs and traditional raw features. The reviewed studies demonstrate that GNN-based models, in particular, provide significant performance improvements over classical methods through automatic feature extraction from both node and edge attributes. Approaches focusing specifically on edge features and utilizing data derived from network flows have shown superior capability in capturing attack patterns, achieving notably high F1-scores. Moreover, the results obtained on various datasets (e.g., NF-UNSW-NB15-v2) support these approaches' practical applicability and effectiveness in real-world scenarios. However, several limitations and areas for improvement remain in the current literature. While feature extraction is recognized as a significant advancement, the full potential of graph structures—especially edge features—has yet to be fully explored.

In conclusion, graph-based IDS methods are making significant contributions to next-generation cyber-attack detection and hold substantial potential for broader and more effective deployment soon.

4.3 | Graph-Based Approaches for Botnet Detection

Botnets are systems in which many devices (bots), controlled from a central entity, conduct cyber-attacks in a coordinated manner. Detecting such threats poses significant challenges for traditional methods, as bots often mimic normal user behavior. Graph-based representations offer a more effective analytical approach by modeling these coordinated and structured behaviors. In graph-based botnet detection studies, nodes represent devices or connections, while edges denote traffic flows. These studies frequently employ algorithms such as Graph Neural Networks (GNNs), Random Forest, and Support Vector Machines (SVMs). Some approaches involve direct feature extraction at the node or edge level. Notably, datasets such as CTU-13, IoT-23, and CICIDS2017 are commonly used across the literature. A list of relevant studies is listed in Table 3.

Wang et al. present a hybrid model developed for botnet detection. This model aims to achieve high accuracy and protocol-independent detection by analyzing flow-based and graph-based network traffic behaviors. The graph analysis performs anomaly detection using the egonet structure on the communication graph. In contrast, k-means clustering and stability measurements are applied in the flow analysis based on 15 statistical features. The model makes decisions through an ensemble method combining similarity, stability, and anomaly scores. The features used in the model include flow-based statistical attributes such as packet duration, count, and sizes, as well as graph-based features like the number of nodes, edges, and total weight within the egonet. The study simulated traffic data from Mirai, Zeus, Athena, Black Energy, and Ares botnets in real environments. Experimental results show that the flow-based detector achieves 99.49% accuracy, the graph-based detector 91.66%, and the hybrid use of both methods reaches a very high accuracy of 99.94%. This approach compensates for the weaknesses of each technique, reducing false favorable rates and increasing effectiveness in botnet detection. Additionally, the BotMark model stands out for its successful results across different botnet types without depending on their command and control (C&C) protocols [79].

In a study by Millar et al., a graph-based machine learning method named 'BiSAGE' was introduced for detecting devices involved in bot networks. In the survey, manual feature extraction was performed for botnet detection, and 10 flow-based features were identified. These features include the average length of packets sent to the destination IP and from the source IP,

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TABLE 3 | A list of research studies using graph-based AI techniques for botnet detection.

References	Overview	Classification methods	Dataset	Year	Domain	Threat type
[79]	We propose "BotMark" for botnet detection through hybrid analysis of flow-based and graph-based network traffic behavior	Similarity-Based Detector, Stability-Based Detector, Graph-Based Detector	Real computing environment	2020	Botnet	Botnet detection
[80]	A graph-based machine learning method named "BiSAGE" is proposed for botnet detection	SVM	CSE-CIC-IDS2018	2021	Botnet	Botnet detection
[81]	BotSward is a proposed graph-based method for detecting botnets	RF, Gradient Boosting (GBC), LR, Stochastic Gradient Descent (SGD), DT, KNN, SVM ve CNN	CTU-13	2023	Botnet	Botnet detection
[82]	A graph is created from network traffic data, and botnet detection is carried out by extracting features using fundamental graph-based metrics	LSTM	CTU-13	2019	Botnet	Botnet detection
[83]	A graph-based machine learning method is proposed for botnet detection	NB, DT, RF, AdaBoost, KNN, ExtraTrees	CTU-13, IoT-23	2021	Botnet	Botnet detection
[84]	A two-stage graph-based bot detection system employing both supervised and unsupervised machine learning algorithms is proposed	K-Means, DBScan, DT, LR, SVM and FNN (Feed-Forward Neural Network)	CTU-13	2019	Botnet	Botnet detection
[85]	A novel graph-based method for detecting botnets is introduced	SVM	CTU-13	2017	Botnet	Botnet detection
[86]	The *E-GraphSAGE* model is developed for IoT intrusion detection by leveraging edge embeddings and flow-based topological features, outperforming traditional ML classifiers with up to 99.99% accuracy.	E-GraphSAGE	BoT-IoT, ToN-IoT, NF-BoT-IoT	2023	Botnet	Botnet detection

packet sending rates, total number of packets and bytes, and the average time intervals between packets. While some studies in the literature combine the "Forward" and "Backward" directions of these features into a single metric, this study avoided such merging to prevent loss of information. Instead, each feature was treated separately to obtain a more detailed and meaningful feature representation. The CSE-CIC IDS2018 dataset was employed in the study, and the SVM algorithm was utilized for classification. The proposed BiSAGE method in the study was evaluated on the CSE-CIC-IDS2018 dataset and tested using a one-class SVM model trained only with benign samples. This approach is significant in terms of its ability to detect unknown botnet behaviors using unlabeled data. The results demonstrate that BiSAGE achieves significantly higher classification performance than existing methods such as GraphSAGE and Node2Vec. Specifically, BiSAGE attained an F1-score of up to 96.1% with a two-level ($k = 2$) neighborhood aggregation structure. This success reveals that botnet victims can be effectively detected based on their behaviors and by considering the behaviors of the devices they communicate with. The study's most important contribution is that it enables high-accuracy detection without requiring labeled botnet data, thereby allowing identification of new and unknown botnet types [80].

In the study by Shinan et al., BotSward, a machine learning-based graph-based bot detection system, was introduced using the CTU-13 dataset. The study converts network traffic data into graphs to facilitate feature extraction, which includes metrics such as proximity, degree centrality, and PageRank values. For classification purposes, the study employs various machine

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learning algorithms, namely Random Forest (RF), Gradient Boosting Classifier (GBC), Logistic Regression (LR), Stochastic Gradient Descent (SGD), Decision Tree (DT), K-Nearest Neighbor (KNN), Support Vector Machine (SVM), and Convolutional Neural Network (CNN). In the study, the proposed method achieved an accuracy rate of up to 99% in experiments conducted on the CTU-13 dataset. BotSward provides a botnet detection solution that is resilient to unknown attack types, scalable, and suitable for real-time applications, thereby making a significant contribution to the literature [81].

Sinha et al. proposed an approach for botnet detection where a graph is constructed from network traffic data. Feature extraction involved basic graph-based metrics such as Out-Degree, In-Degree, Out-Neighbors, In-Neighbors, PageRank centrality, Betweenness centrality, Eigenvector centrality, and Local Clustering Coefficient. The proposed method in the study offers a content-independent and generalizable approach for detecting botnet-infected devices by monitoring the temporal evolution of communication structures within the network. By modeling graph-based features as time series and classifying them using an LSTM (Long Short-Term Memory) neural network, high generalizability against various botnet types has been achieved. Evaluations conducted on five different scenarios of the CTU-13 dataset yielded successful results with 96.2% accuracy, 94.6% sensitivity (TPR), and 96.3% specificity (TNR). Additionally, the model demonstrated a high area under the ROC curve (AUROC), indicating strong generalization capability within and across scenarios. Compared to content-dependent methods in the literature, this approach—which operates without relying on botnet protocols, port numbers, or packet contents—achieves the highest success among content-independent methods. In this regard, the study makes a significant contribution to the literature by providing an effective solution capable of real-time detection of hard-to-detect botnet behaviors [82].

Alharbi et al. developed a graph-based machine learning model for botnet detection, which involves creating a graph representation from network traffic data, followed by feature extraction and normalization operations to prepare input for the machine learning algorithms. Feature extraction includes metrics such as in-degree, out-degree, in-degree weight, out-degree weight, clustering coefficient, in-degree centrality, out-degree centrality, node betweenness, authority, hub, Katz centrality, PageRank, closeness centrality, and eigenvector centrality. The study evaluates several machine learning algorithms, including Naive Bayes (NB), Decision Tree (DT), Random Forest (RF), AdaBoost, K-Nearest Neighbors (KNN), and ExtraTrees on datasets such as CTU-13 and IoT-23. Results indicate that the proposed approach reduces training time, simplifies model complexity, and achieves high bot detection rates. This study's graph-based machine learning model was tested on real botnet traffic datasets such as CTU-13 and IoT-23, demonstrating high accuracy and generalizability. In particular, when using the ExtraTrees classifier combined with feature selection based on Pearson correlation, the model achieved superior performance with 99%–100% accuracy. Additionally, the model showed resilience against zero-day attacks and successfully detected botnet activities at an early stage. Comparative evaluations showed that the proposed model outperformed graph-based methods like Botchase and flow-based contemporary approaches like BotFP-Clus. Compared to two-layer detection systems, it offers lower processing time and higher precision, providing an effective, real-time, and scalable botnet detection solution. In this regard, the study makes a significant contribution to developing powerful security models for emerging technologies such as IoT and cloud systems [83].

Daya et al. proposed a two-stage graph-based bot detection system utilizing supervised and unsupervised machine learning algorithms. The study begins by converting network traffic into a graph representation, followed by feature extraction and normalization. These features are input to supervised and unsupervised algorithms in the final stage. Feature extraction includes metrics such as in-degree, out-degree, in-degree weight, out-degree weight, betweenness centrality, local clustering coefficient, and alpha centrality. The study utilized the CTU-13 dataset and employed algorithms such as k-Means, DBScan, Self-Organizing Maps (SOM), Decision Trees (DT), Logistic Regression (LR), Support Vector Machines (SVM), and Feed-forward Neural Networks (FNN). According to the findings, the proposed system demonstrates robustness against zero-day attacks. The two-stage graph-based machine learning approach proposed in this study combines unsupervised learning (SOM) and supervised learning (DT) to provide an effective solution for botnet detection with high accuracy and a low false positive rate. Experiments conducted on the CTU-13 dataset yielded notable results with 90.9% accuracy. Specifically, in the first stage, the SOM algorithm was used to filter out benign nodes from the large dataset, enabling the second stage to achieve high performance with a decision tree (DT) on a smaller, more focused dataset. Additionally, the proposed system demonstrated resilience against zero-day attacks and produced consistent results across different network topologies. Feature normalization ensured structural similarity among nodes from different datasets, enhancing the model's spatial stability. In this regard, the study offers a robust and generalizable solution for real-time, protocol-independent botnet detection in large-scale networks [84].

In the study performed by Chowdhury et al., a novel graph-based method for detecting botnets was introduced, emphasizing its rule-independent capability to capture evolving bot behaviors. The technique employed metrics such as in-degree, out-degree, sum of ingoing edge weight, sum of outgoing edge weight, clustering coefficient, node betweenness, and eigenvector centrality for feature extraction. The research utilized the CTU-13 dataset and employed the SVM algorithm for classification. Using seven different graph features (in-degree, out-degree, in/out-degree weight, clustering coefficient, betweenness centrality, eigenvector centrality), nodes were clustered with the Self-Organizing Map (SOM) algorithm, isolating bots in small-sized clusters. Consequently, the number of nodes to be examined for bot detection was reduced to below 0.1%, significantly decreasing computational cost. Furthermore, filtering out low-degree (single-connection) nodes further reduced the number of nodes within bot clusters. Compared to classification-based methods such as SVM, the proposed method was more resilient to variability in bot behaviors and successfully detected different botnet types [85].

Weng et al. proposed a graph neural network (GNN) approach that uses flow-based network data for anomaly detection in

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IoT networks. A new GNN model called E-GraphSAGE was developed, capable of capturing both edge features and the topological structure of the network traffic graph. The model was tested on recent datasets such as BoT-IoT, ToN-IoT, NF-ToN-IoT, and NF-BoT-IoT. The results demonstrated that the E-GraphSAGE-based network intrusion detection system outperformed traditional classifiers, with accuracy rates ranging from 93% to 99.99%. Thanks to the edge embedding approach, E-GraphSAGE could accurately distinguish attack traffic from regular traffic, and visualization analyses performed with UMAP highlighted this separation. In this regard, the study illustrates the potential of GNN architectures for real-time and generalizable attack detection in IoT networks, making a significant contribution to the literature [86].

Studies on botnet detection have demonstrated high accuracy by revealing the coordinated communication structures of botnets through topological, graph-based analysis. While the CTU-13 dataset is a standard benchmark in nearly all studies, high performance—ranging between 96% and 99% accuracy—has also been reported on alternative datasets such as CSE-CIC-IDS2018 and IoT-23. These results have been achieved using deep learning methods (e.g., Graph Neural Networks) and traditional machine learning algorithms (e.g., Support Vector Machines and Random Forest). However, the evolving nature of botnets and the challenge of detecting multiple botnet types simultaneously highlight the growing need for graph models that support real-time updates and transfer learning capabilities. Addressing these aspects is crucial for ensuring the robustness and adaptability of detection systems in dynamic threat environments.

4.4 | Graph-Based Approaches for Detecting IoT and DDoS-Based Attacks

With the rapid proliferation of IoT devices, their inherent security vulnerabilities have made them attractive targets for cyber attackers. The complex, irregular, and often dynamic traffic patterns generated by these devices present a broad attack surface that is difficult to analyze using traditional methods. Similarly, DDoS attacks, characterized by high-volume and simultaneous traffic behavior, can easily mislead conventional detection systems. Graph-based representations offer a robust foundation for modeling inter-device relationships, traffic intensity, and topological characteristics in such scenarios. Studies in this domain typically represent network traffic as graphs, where nodes denote devices or sessions, and edges correspond to communications or data flows. Detection systems have been built using Graph Neural Networks (GNNs), hybrid models such as CNN-LSTM, and various feature extraction strategies. Commonly used datasets in these studies include BoT-IoT, ToN-IoT, NF-ToT-IoT, and NF-BoT-IoT. Related studies are outlined in Table 4.

Khalaf et al. proposed a multi-agent distributed system for anomaly detection, introducing the Graph-Based Layered Learning Driven (GLLD) model. The proposed architecture comprises decentralized agents, where each node learns anomaly patterns by interacting only with its immediate neighbors in a graph. This communication-aware structure eliminates the need for centralized processing and significantly reduces bandwidth and power consumption, making it ideal for resource-constrained IoT environments. The model was evaluated using two distinct datasets: BoT-IoT (representing flooding and DoS attacks in simulated IoT scenarios) and the Code Red worm attack dataset (a real-world malware scenario). Feature extraction included parameters such as start time, duration, protocol, source/destination devices, connection direction, number of packets, and transmitted bytes, allowing for detailed network behavior modeling. The performance results demonstrate the superiority of the GLLD approach: the system achieved ROC-AUC scores between 98.4% and 99.3%, with very low false favorable rates and minimal latency. These outcomes confirm the method's robustness in accurately detecting the complexity of inter-node dependencies, especially under distributed and evolving attack conditions [87].

In the study conducted by Paudel et al., a real-time and encryption-agnostic graph-based model called GODIT (Graph-based Outlier Detection in IoT) was proposed to detect DoS attacks occurring in smart home IoT devices. In the model, network traffic was represented as a real-time undirected graph stream where communications between devices are modeled as nodes and edges, with each minute of traffic processed as a separate graph. Since the model learns solely from source and destination IP information without requiring additional traffic features (such as packet size, protocol, port number, etc.), it is particularly effective in encrypted network environments. Tested in a real IoT testbed environment (UNSW Smart Home Testbed) using data collected from 28 different devices, GODIT achieved an F1 score of 92% with a Decision Tree in a static scenario, and demonstrated significant superiority over other approaches in the flow scenario with more than 7000 graphs, achieving 66% recall and 67% precision [88].

The study presented by Shih et al. proposes an innovative and lightweight approach for detecting DDoS attacks directly from the graph structure without any feature extraction. The model represents network traffic as a directed graph where IP addresses are modeled as nodes and data transmissions between them as edges. The graph structure is continuously updated within time windows, allowing real-time analysis of topological changes caused by high-volume attacks such as SYN Flood and DNS Flood. This method does not require any statistical feature engineering or protocol-based content analysis; it relies solely on the density of connections between nodes and the directions of interactions. The most striking aspect of the model is its ability to detect attack moments directly from patterns reflected in the graph structure of all packets, without any manual or statistical feature extraction. This approach offers a significant advantage, particularly for analyzing encrypted or metadata-limited traffic. Tested on real traffic scenarios, the system achieved accuracy rates exceeding 99% and detected attack onsets within milliseconds. Additionally, its low computational overhead stands out as a highly applicable solution for real-time systems. In this context, the study can be considered a pioneering example in graph-based DDoS detection that does not rely on feature engineering [89].

The study conducted by Protogerou et al. proposes a novel distributed anomaly detection approach tailored for IoT networks using Graph Neural Networks (GNNs) within a multi-agent system (MAS) framework. The method models IoT network

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TABLE 4 | A list of original research studies employing graph-based AI methods for IoT/DDoS attack detection.

References	Overview	Classification methods	Dataset	Year	Domain	Threat type
[87]	The Graph-based Layered Learning-Driven method proposes feature extraction by creating a graph.	SVM, DT, RF GLLD (Graph-based Layered Learning-Driven)	BoT-IoT, Code Red	2022	IoT/DDoS	Flooding, DoS
[88]	A new approach called Graph-based Outlier Detection in Internet of Things (GODIT) is proposed to represent smart home IoT traffic as a graph flow and detect real-time DDoS attacks.	Decision Tree, Support Vector Machine, Gradient Boosting, Random Forest	UNSW Sydney	2019	IoT/DDoS	DDoS detection
[89]	A real-time DDoS detection approach is proposed to model traffic as a directed IP graph and identify attack bursts purely from graph structure without any feature engineering.	Graph Structure Analysis	Real Network Traffic	2020	IoT/DDoS	DDoS detection
[90]	Using localized communication graphs, a distributed GNN-based anomaly detection system within a Multi-Agent System (MAS) framework is introduced to identify DDoS and worm attacks across IoT networks.	GNN in Multi-Agent Systems	CTU-13	2022	IoT/DDoS	DDoS detection
[91]	The *Wrongdoing Monitor* framework constructs Behavioral Identification Graphs (BIG) to model event-property associations and detect anomalies across fraud, insider threats, and intrusion domains via graph embedding.	Behavioral Identification Graph (BIG)	Real Network Traffic	2023	Behavioral Anomaly	Behavioral anomaly
[92]	A graph model is employed to represent the structural characteristics of traffic data between targeted systems and bot devices.	PCA + FCM	CIC-IDS-2017	2022	IoT/DDoS	DDoS detection
[93]	A novel machine learning approach is proposed for detecting DDoS attacks.	Fuzzy C-Means (FCM), PCA-FCM, K-Means, PCA-K-means, (Nonnegative Matrix Factorization) NMF-K-Means	CICIDS-2017	2022	IoT/DDoS	DDoS detection

traffic as graphs where nodes represent devices or forwarding nodes and edges represent communication links, incorporating structural and temporal features extracted from raw network flows. By deploying intelligent agents on active nodes such as IoT devices, SDN forwarders, or Fog nodes, the system enables localized anomaly detection through GNNs that aggregate and transform features from neighboring nodes and edges. This distributed setup not only enhances detection accuracy, particularly against propagating attacks like DDoS and worm infiltration, but also reduces resource consumption by limiting data exchange to neighboring nodes rather than a central server. Feature extraction converts flow-level data into node and edge-level attributes, including average packets sent/received, bytes sent/received, and connection duration, forming feature vectors updated iteratively via multi-layer perceptrons (MLPs) embedded in the GNN. The model is trained and evaluated on real-world (CTU-13) and synthetically generated datasets simulating various attack scenarios. Experimental results demonstrate that the proposed GNN-based MAS outperforms traditional centralized machine learning classifiers (SVM, Decision Tree, Random Forest) and state-of-the-art deep learning methods in detection accuracy and robustness, even under compromised nodes or offline conditions. Additionally, the distributed approach significantly lowers bandwidth and power consumption, addressing critical constraints in IoT

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environments. The study discusses applications across diverse IoT domains, including smart grids, medical IoT, food chain infrastructure, intelligent transportation systems, and computer vision, highlighting the method's scalability and adaptability. Future work aims to incorporate temporal graph models (graph-RNNs), enhance multi-hop neighborhood views, and integrate the approach with Software Defined Perimeter architectures for improved security [90].

The study conducted by Wang et al. introduces Wrongdoing Monitor, a novel graph-based behavioral anomaly detection (BAD) framework designed to improve cybersecurity across multiple domains. The Behavioral Identification Graph (BIG) is central to the approach, which models behavioral data at two levels: fine-grained property-level co-occurrence associations and coarse-grained event-level relationships. This unified graph and spatial representation captures deep inter- and intra-behavioral correlations that existing methods often overlook. The BIG framework constructs a property graph from behavioral event data, representing each property as a node and the co-occurrences as weighted edges. The system effectively learns latent embeddings that enhance anomaly detection by integrating network representation learning techniques with an event-property composite model. The model leverages temporal dynamics via the Hawkes process to weigh edge significance over time. The approach was validated on diverse datasets representing fraud detection in online payments, intrusion detection in network communication, insider threat detection, and compromise detection in social networks. Experimental results demonstrate superior performance to state-of-the-art baseline models, achieving higher detection rates with reduced false positives. The method is compatible with various network representation learning algorithms and adaptable across different anomaly detection scenarios. This work highlights the effectiveness of integrating multi-level behavioral associations within graph structures to enhance anomaly detection, offering a scalable and generalizable solution for real-world cybersecurity challenges [91].

In the study by Jing et al., a novel method for DDoS detection is proposed by leveraging graph structural features combined with Non-negative Matrix Factorization (NMF). The network traffic is modeled as a directed graph where nodes represent IP-port pairs and edges denote communication flows. Eight key graph and flow-based features, such as in-degree, out-degree, eigenvector centrality, clustering coefficient, flow duration, and packet size variability, are selected to characterize communication patterns during DDoS attacks. The method applies Principal Component Analysis (PCA) for dimensionality reduction, followed by Fuzzy C-Means (FCM) clustering to classify network flows without requiring labeled data, enabling unsupervised detection of attacks. Evaluation on the CICIDS-2017 dataset demonstrates high detection accuracy and low false positive rates. This approach highlights the effectiveness of combining graph theory with matrix factorization and clustering techniques for detecting sophisticated network anomalies [92].

In the study by Jing et al., a machine learning approach for detecting DDoS attacks in Industrial IoT devices is introduced by analyzing network traffic graph structure features. Eight traffic-related features extracted from edge weights are refined using Principal Component Analysis (PCA). Fuzzy C-Means (FCM) clustering is applied to separate normal and attack traffic without labeled data. Although the article was later retracted due to peer-review issues, the approach illustrates the potential of graph-based traffic representation combined with clustering for unsupervised anomaly detection in IoT networks. The method addresses challenges in real-time DDoS attack identification by modeling communication topology and traffic flow metrics [93].

Graph-based approaches for DDoS attack detection offer a level of representational power that goes beyond traditional methods by modeling the structural characteristics of network traffic. The reviewed studies are generally based on frameworks that model communication flows using IP-based directed graphs or node-edge graph structures. Most of these studies demonstrate that GNN-based models — such as GraphSAGE, GCN, GAT, and ResGAT — successfully distinguish malicious traffic by effectively processing both node and edge features. As a result, graph-based DDoS detection research contributes substantially to the field, particularly through its ability to capture structural patterns and flexible modeling capabilities.

4.5 | Semantic Behavior-Based Attacks: Detection of Android Malware and APTs

Malware and Advanced Persistent Threats (APTs) primarily rely on in-system behaviors, semantic relationships, and temporal patterns, rather than conventional network-level activity. In particular, threats such as Android malware and APT attacks cannot be fully understood through superficial traffic analysis alone; instead, their detection requires modeling in-app behaviors, call chains, and cause-and-effect relationships. Consequently, recent research has increasingly adopted methods such as graph representation, time series modeling, and embedded behavioral patterns to detect such sophisticated threats. Graph-based representations enable the abstract yet meaningful capture of attack patterns by modeling function calls, instruction sequences, API relationships, and semantic contexts through nodes and edges. In the Android ecosystem, malicious behavior is often implemented through abuse of system permissions, suspicious API calls, or manipulation of control flow. Therefore, graph types such as Function Call Graphs (FCGs), behavioral graphs, and control flow graphs are commonly employed in malware analysis to uncover underlying threat structures.

In 2020, the study conducted by Niu et al. (2020) proposes a novel Android malware classification method based on deep learning applied to OpCode-level Function Call Graphs (FCG). Given the rising prevalence of Android-based IoT devices and the increasing threat from malware targeting these platforms, the authors focus on enhancing malware detection accuracy by leveraging static analysis of OpCode sequences derived from FCGs. Their approach extracts the Function Call Graph from the Dalvik bytecode of Android applications using the FlowDroid tool and disassembly techniques. OpCode sequences are generated by traversing the FCG, capturing fine-grained behavioral characteristics of malware that traditional signature-based or simplistic feature-based methods often miss. They employ a Long Short-Term Memory (LSTM) neural network to model temporal and semantic dependencies within OpCode sequences for classification. Experiments were conducted on a dataset of

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1796 malicious Android samples (categorized mainly as Trojans and Adware) and 1000 benign applications collected from VirusShare, AndroZoo, and commercial app stores. The method achieved an accuracy of 97%, outperforming other state-of-the-art approaches while reducing processing time. This study highlights the effectiveness of combining program structure analysis (via FCG) with deep learning models to detect Android malware with high precision and robustness, providing a promising solution for security in Android and IoT ecosystems [94].

In 2023, the study conducted by Wu et al. proposes DeepCatra, a multi-view deep learning framework for Android malware detection. The model integrates a bidirectional LSTM (BiLSTM) and a Graph Neural Network (GNN) to leverage temporal opcode sequences and graph-structured flow information derived from static call traces targeting critical APIs identified via vulnerability repositories. DeepCatra constructs a static call graph enriched with inter-component communication edges and extracts call traces ending at critical API calls. The BiLSTM subnet processes opcode sequences sampled along these call traces to capture temporal execution features. In contrast, the GNN subnet embeds an abstract flow graph representing the relationships between critical call traces and inter-component flows, including intent communications. Evaluated on a large dataset of over 18,000 real-world benign and malicious apps, DeepCatra outperforms existing deep learning approaches such as CNN, LSTM, and GCN in detection accuracy and F1 score, achieving improvements between 2.7% and 14.6%. The strategy is robust against code obfuscation and scalable, benefiting from static analysis that avoids the overhead and incompleteness of dynamic profiling. The authors highlight that incorporating critical API knowledge and heterogeneous flow types into the graph embedding enhances malware behavior representation, making DeepCatra a practical and effective solution for Android malware detection [95].

The study conducted by Cuzzocrea et al. introduces a graph-representation-learning framework aimed at enhancing Android malware identification and addressing polymorphic malware evolution. The approach's core is the Inferential Structural Iterative Representation Learning method for Graph Nodes (Inferential SIR-GN), which preserves graph structural similarities through iterative clustering and aggregation. Evaluated on the large-scale MALNET-TINY dataset (over 61,000 samples with 43 malware types and 246 families), the proposed method achieved a macro-F1 score of 71.8%, macro-precision of 72.9%, and macro-recall of 79.4% on a random split. The model reached even higher scores on a smaller subset, MALNET-TINY 5 k with a macro-F1 of 91.6%, precision of 91.7%, and recall of 91.5%. Temporal split experiments showed a macro-F1 of 72.5%, precision of 73.9%, and recall of 80.7%, indicating good generalization to future unseen malware. The method outperformed ResNet18, a neural network baseline for image-based malware classification, which scored a macro-F1 of 65.1% on the same tasks. This framework also incorporates polymorphic malware evolution modeling, improving detection robustness against obfuscated variants. Overall, the approach demonstrates superior accuracy and adaptability for large-scale Android malware classification [96].

The study by Abdelrahman Elsharif Karrar (2022) presents a graph-based machine learning approach for Android malware detection, focusing on API calls and permissions extracted from a dataset of 15,036 applications from the Drebin project. The approach constructs connected flow graphs to represent app features and applies feature selection techniques to identify critical API calls and permissions for malware classification. Four machine learning classifiers—Logistic Regression (LR), Decision Tree (DT), K-Nearest Neighbors (KNN), and Random Forest (RF)—are trained and evaluated using both 66% data splits and ten-fold cross-validation methods. Among these, the Random Forest classifier achieves the highest accuracy of 97.3% and recall of 96.8%, while KNN and DT classifiers reach around 96% accuracy. Logistic Regression shows slightly lower performance with approximately 95% accuracy. Feature selection using ANOVA and SelectKBest filters highlights key API calls and permissions significantly contributing to malware detection. The study also analyzes processing times, showing Random Forest to be the most efficient among the tested classifiers, requiring less than ten seconds on average. This research underscores the effectiveness of graph-based representations combined with machine learning classifiers for high-accuracy and resource-efficient Android malware detection. Future work is suggested to explore dynamic feature extraction and evaluate performance on larger, more diverse datasets to enhance privacy protection and detection robustness [97]. The study by Xin et al. (2018) provides a comprehensive review of machine learning (ML) and deep learning (DL) methods applied to cybersecurity, particularly focusing on network intrusion detection systems (IDS). The article categorizes and explains popular ML techniques including Support Vector Machines (SVM), k-Nearest Neighbors (kNN), Decision Trees (DT), and ensemble methods, as well as DL models like Deep Belief Networks (DBN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Convolutional Neural Networks (CNN). Widely used benchmark datasets such as KDD-Cup 99, NSL-KDD, DARPA, and ADFA are described, with their characteristics and common attack types detailed. The authors discuss the strengths and limitations of each ML and DL technique in terms of detection accuracy, false alarm rates, interpretability, and computational requirements. Performance metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC, are explained, highlighting their importance in evaluating IDS effectiveness. The review emphasizes challenges such as data imbalance, feature selection, novel (zero-day) attack detection, and deployment efficiency. Finally, the article outlines future directions in cybersecurity ML/DL research, advocating for hybrid models combining ML and DL, improving detection speed, online learning for adapting to evolving threats, and developing more comprehensive and balanced datasets [98].

These studies demonstrate that graph-based relationships at the behavioral level are particularly effective in detecting malware and offer detailed analytical capabilities beyond traditional methods. APT attacks, on the other hand, typically involve threats that persist within the target system for extended periods while carrying out low-profile actions based on complex scenarios. To uncover such threats, it is essential to model the temporal relationships between events, cause-effect chains, and system behavior networks using graph representations. The study conducted by Li et al. proposes a hierarchical approach for Advanced Persistent Threat (APT) detection using novel attention-based Graph

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Neural Networks (GNNs). To comprehensively capture behaviors across the full APT lifecycle, the authors construct two complementary graphs: the Intrahost Provenance Graph (IPG) representing system-level events, and the Interhost Interactive Graph (IIG) capturing network-level interactions. A metapath aggregated GNN is applied to the heterogeneous IPG to embed complex semantic and structural system behaviors. At the same time, an edge feature-enhanced GNN is designed for the IIG to model rich, multidimensional edge features reflecting host interactions. The framework includes a dynamic enhancement mechanism that updates the IIG detection model based on suspicious hosts identified by the IPG detectors, enabling continuous learning of new attack patterns. Evaluation on two benchmark datasets—StreamSpot (system-level) and LANL (network-level)—demonstrates the approach's effectiveness. On the StreamSpot dataset, the IPG detector achieves superior performance with an accuracy and F1-score of 98%, outperforming baseline methods by over 10%–25%. The IPG detector also reaches an AUC of 0.98, with an actual positive rate (TPR) of 0.98 at a false positive rate (FPR) of 0.1. On the LANL dataset, the IIG detector attains an AUC of 0.90, with precision, recall, and F1-score all at 83%, surpassing competing methods such as Tiresias, log2vec, and GAT. The dynamic model update mechanism further improves performance, achieving an AUC of 0.95 and TPR of 0.92 when incorporating reported malicious hosts. The hierarchical detection framework effectively models APT behaviors at both system and network levels, enabling robust and scalable anomaly detection in complex environments [99].

The study conducted by Fu et al. (2022) proposes ST-Graph, a real-time encrypted malware traffic detection framework based on graph representation learning. ST-Graph constructs a heterogeneous spatio-temporal graph to model encrypted TLS network traffic by integrating spatial connection patterns and temporal communication sequences among hosts and servers. The method employs a novel interval-inclined random walk algorithm to learn edge embeddings that capture dynamic network behavior efficiently. Evaluated on two datasets—EncMal2021 (a large-scale dataset with over 100,000 hosts and 5 million streams) and AndMal2019 (a public dataset with benign and malicious Android traffic)—ST-Graph demonstrates outstanding detection performance. On EncMal2021, ST-Graph achieves 99.98% precision, 99.92% recall, and a remarkably low 0.0045% false positive rate (FPR) in malware detection, outperforming baseline models ETA and FS-Net. It also attains 93.37% precision and 91.21% recall in malware family classification. On AndMal2019, ST-Graph attains 99.30% precision, 99.64% recall, and 0.017% FPR for malware detection, significantly surpassing other methods. The framework maintains high precision and recall even on disjoint testing sets with unseen malware families, showing excellent generalization. Real-world deployment over nearly a year in campus and enterprise networks confirms its efficiency, processing multi-Gbps traffic with low latency and maintaining superior detection accuracy. This study highlights the advantage of integrating multi-feature spatial–temporal graph embeddings for detecting sophisticated encrypted malware traffic in large-scale, real-world networks [100].

The study by Liu et al. proposes a novel causal graph-based framework for predictive analytics of multi-stage Advanced Persistent Threat (APT) attacks using audit log data. The approach constructs platform-independent causal graphs representing causal relationships among system entities and events, which are then reduced through causality-preserving pruning methods to ensure computational efficiency while maintaining semantic integrity. Two types of attack scenario graphs are extracted—T-evolving graphs and neighborhood graphs—capturing temporal and contextual attack event sequences. These sequences are converted into vector representations via word embedding and modeled with a hybrid BiLSTM-CNN deep learning architecture to learn and predict subsequent malicious events. Evaluated on realistic datasets simulating multiple APT campaigns derived from DARPA Transparent Computing audit logs, the method demonstrates high accuracy and efficient real-time predictive capability, effectively forecasting forthcoming attack stages. This system aids network analysts by recommending probable next malicious events with associated probabilities, enhancing proactive cyber defense [101].

In the detection of advanced cyber threats—such as Android malware, APT attacks, and encrypted traffic-based intrusions—graph-based approaches offer superior accuracy, generalizability, and explainability compared to traditional content-driven methods. In the reviewed Android-focused studies, particularly high accuracy rates (up to 97%) have been achieved by integrating flows extracted from OpCode-level Function Call Graphs (FCGs) with sequential models such as LSTM or BiLSTM. In APT detection, representations such as provenance graphs and causal graphs, which reflect cause-and-effect chains between events, have been employed to detect multi-stage attack patterns with accuracies ranging from 85% to 98%. For encrypted traffic analysis, methods like ST-Graph have achieved over 99% accuracy even in modern protocols (e.g., TLS 1.3) and obfuscated traffic scenarios, by learning edge embeddings over spatio-temporal integrated heterogeneous graph structures. Representative works are listed in Table 5.

Furthermore, tests conducted on large-scale datasets demonstrate that these approaches are effective, scalable, and suitable for real-time system integration. In conclusion, graph-based learning goes beyond conventional feature extraction by capturing multi-dimensional, temporal, and relational patterns, offering a robust paradigm for the early and accurate detection of cyber threats. This review has systematically examined the integration of graph-based representation techniques in AI-assisted cyber-attack detection. The literature reveals that while traditional machine learning and deep learning approaches have shown high classification accuracy, they often fall short in capturing the complex structural and relational characteristics inherent in network traffic. Graph-based methods offer a promising alternative by effectively modeling the interconnectivity and temporal dynamics of cyber events. Across various subdomains such as general IDS/NIDS systems, botnet detection, IoT/DDoS attacks, and semantic behavior-based threats (e.g., Android malware and APTs), researchers have leveraged graph representations in diverse ways: either through direct learning on graph structures using GNNs or by extracting network-theoretic features for traditional classifiers. Notably, feature extraction from graph representations has also proven valuable across domains beyond cybersecurity, further validating its methodological robustness.

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TABLE 5 | A list of research studies using graph-based AI methods for detecting android malware and APT attacks.

References	Overview	Classification Methods	Dataset	Year	Domain	Threat type
[94]	A deep learning-based method is proposed for Android malware classification using OpCode-level Function Call Graphs (FCGs) and LSTM models to capture temporal dependencies in execution patterns.	LSTM on OpCode-FCG	VirusShare, AndroZoo	2020	Malware	Android malware
[95]	The *DeepCatra* framework integrates BiLSTM and GNN to jointly learn from temporal OpCode sequences and static call graph structures enriched with inter-component flows for Android malware detection	BiLSTM + GNN (DeepCatra)	Real benign/ malicious apps	2023	Malware	Android malware
[96]	A graph-based technique is employed to extract control flow graphs from Android APK binary files. This approach utilizes a new graph representation called SIR-GN, which preserves graph structural similarities.	XGBoost	MALNET	2023	Malware	Android malware
[97]	A graph-based machine learning algorithm is proposed for managing mandatory permissions and API functions, using application data from the Drebin project to identify the most crucial features for malware detection.	RF, KNN, DT, LR	Derbin	2022	Malware	Android malware
[98]	A comprehensive review of machine learning and deep learning techniques for intrusion detection systems (IDS) is presented, categorizing popular models, datasets, and performance metrics while highlighting challenges such as data imbalance, interpretability, and real-time applicability.	ML (SVM, kNN, DT, Ensemble), DL (DBN, RNN, LSTM, CNN)	KDD-Cup 99, NSL-KDD, DARPA, ADFA	2018	Malware	Zero-day attacks
[99]	A hierarchical GNN framework is proposed for APT detection, combining system-level Intrahost Provenance Graphs and network-level Interhost Graphs with attention-based graph embeddings and dynamic learning.	GNN	StreamSpot, LANL	2022	Malware	APT
[100]	A method named ST-Graph is proposed to explore the spatial and temporal properties of network behavior using a graph representation learning algorithm.	ETA, FS-Net, ST-Graph	CICInvesAnd Mal2019	2022	Malware	APT
[101]	A combination of causal graphs and deep learning techniques is applied for predictive analysis of Advanced Persistent Threats (APTs).	LSTM, CNN + CCR (Causal Conv. Residual), (Transformer- Encoder)	Real Computing Environment	2023	Malware	APT

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The proposed classification framework reveals that graph-based cyber-attack detection approaches are undergoing a clear developmental trajectory in terms of both application diversity and methodological maturity. However, some fundamental challenges remain unaddressed in the field. In particular, the lack of publicly available datasets directly suitable for graph representation, standardized, and capable of supporting comparative experiments constitutes a significant limitation. Most of the reviewed studies construct graph structures from raw data, and this process typically relies on customized preprocessing steps. Such variability weakens methodological consistency across studies and undermines the reliability of performance comparisons. Therefore, future research should prioritize the development of open datasets suitable for graph-based modeling and the design of explainable, scalable, and real-time applicable graph models based on these datasets.

5 | Discussion and Key Insights

This review systematically analyzes graph-based approaches for AI-assisted cyber-attack detection, offering a domain-oriented classification rather than a purely algorithmic or feature-based taxonomy. Unlike previous studies that group research by algorithm type or feature extraction strategy, this work classifies the literature into four key application areas:

1. General IDS/NIDS systems
2. Botnet and DDoS detection
3. IoT-based threats
4. Advanced threats like Android malware, APTs, and encrypted traffic.

This structure clarifies how graph modeling strategies, learning paradigms, and feature engineering approaches differ depending on the specific threat context.

5.1 | IDS/NIDS Systems

Most studies in this category aim to convert network traffic or flow data into communication graphs or time-series-based visibility graphs (VG/NVG). Traditional approaches rely on graph-theoretic features (e.g., degree and centrality) and feed them into classical machine learning algorithms. However, more recent works directly employ CNN or GNN architectures on visibility graphs, resulting in improved performance with reduced preprocessing.

5.2 | Botnet and DDoS Detection

Graph-based modeling plays a critical role in detecting distributed attacks. Communication graphs constructed from IP flow data are used to capture suspicious patterns. Some studies extract manual features from these graphs, while others apply GNN architectures for direct anomaly detection. Hybrid systems have also emerged, aiming to offer stronger real-time detection capabilities in dynamic environments.

5.3 | IoT-Based Threats

The heterogeneity and resource constraints of IoT devices make detection challenging. Studies in this domain often model system behavior using visibility graphs or protocol-specific interaction graphs. While lightweight, feature-driven classifiers remain common, several studies demonstrate the success of simplified GNN models. These findings reflect the ongoing trade-off between computational cost and detection performance in IoT settings.

5.4 | Advanced Threats (Android, APT, and Encrypted Traffic)

Graph representations are increasingly utilized to analyze sophisticated threats such as polymorphic malware or encrypted traffic. Android-focused studies commonly employ Function Call Graphs (FCG) and OpCode sequences derived through static analysis, which are then processed with deep learning models such as GCNs or LSTMs. Graph mining techniques in complex scenarios like APTs or encrypted communication help uncover behavioral patterns that traditional methods may miss. GNN-based architectures are preferred due to their ability to capture semantic and structural characteristics.

Beyond the distinct characteristics of each application domain, several cross-cutting themes emerge across the literature. These recurring observations shed light on broader methodological trends, practical challenges, and emerging priorities in graph-based cyber-attack detection. The following section synthesizes these insights, offering a comparative perspective that transcends domain-specific implementations.

1. Manual vs. Automated Feature Extraction: While handcrafted graph-theoretic metrics are still relevant—especially in explainability-driven or resource-constrained scenarios—there is a clear shift toward automated representation learning using GNNs and deep learning architectures.
2. Dataset Limitations: There is a notable scarcity of standardized, open-access datasets in areas such as encrypted traffic and APT detection, which limits reproducibility and benchmarking.
3. Operational Challenges: Real-time applicability, explainability, and scalability remain open problems. Although lightweight GNN models and hybrid frameworks show promise, they require further evaluation in dynamic and production-grade environments.

In practice, several fundamental challenges remain unresolved when deploying graph-based methods in real-world cybersecurity systems.

Scalability: As the size of the monitored network grows, the number of nodes and edges in the constructed graphs increases exponentially. This leads to significant memory consumption and computational overhead, especially in GNN-based architectures where message passing and neighborhood aggregation must be computed iteratively. For enterprise-scale IDS/NIDS

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deployments, graphs with millions of flows or connections can drastically increase training and inference times, making the models difficult to scale in production environments. Although lightweight GNNs and sampling strategies have been proposed, their effectiveness under continuous, high-volume network streams is still limited and requires further benchmarking.

Data sparsity and class imbalance: Real-world network traffic is dominated by benign flows, while malicious activities often represent only a tiny fraction of the dataset. This class imbalance problem reduces the sensitivity of models to rare but critical attack patterns. Moreover, in IoT and APT detection scenarios, attack instances are not only rare but also heterogeneous, making generalization even harder. While data augmentation techniques (e.g., SMOTE or synthetic graph generation) and anomaly detection frameworks provide partial remedies, their reliability in large-scale operational environments remains an open question.

Time complexity and real-time applicability: The construction of visibility graphs (VG/NVG) or graph mining processes often introduces substantial preprocessing delays. For example, generating visibility graphs from high-frequency time series data can be computationally expensive, creating latency that undermines real-time detection requirements. Similarly, complex GNN architectures may provide higher accuracy but at the expense of inference latency, which is unacceptable in mission-critical systems that require decisions within milliseconds. Research on streaming graph learning and online adaptation is promising but still in its infancy.

Collectively, these limitations illustrate a persistent gap between experimental research and operational deployment. Addressing scalability bottlenecks, mitigating data sparsity, and reducing computational overhead are therefore critical priorities for future studies. Overcoming these challenges will increase the potential of these models to become more widely applicable solutions in production environments. This study offers a holistic view of the evolution of graph-based cyber-attack detection by combining domain-specific insights with methodological analysis. Future research should prioritize scalable, interpretable, and domain-optimized graph models to advance academic innovation and practical deployment in cybersecurity systems.

6 | Conclusions

This systematic review has explored the growing integration of graph-based techniques within AI-assisted cyber-attack detection. This study provides a more contextualized understanding of how graph modeling methods are utilized in practice by shifting the classification focus from algorithmic categories to application domains. The reviewed literature was grouped into four categories: general IDS/NIDS systems, botnet and DDoS detection, IoT-related threats, and advanced attack scenarios such as Android malware, APTs, and encrypted traffic. The analysis highlighted the variety of graph construction strategies, feature extraction practices, and machine/deep learning models employed within each domain.

A key contribution of this work lies in its distinction between studies that rely on handcrafted graph-theoretic features and those leveraging automated representation learning through Graph Neural Networks (GNNs). This dual perspective enables a deeper understanding of methodological trends and their impact on detection performance. Additionally, the proposed conceptual framework mapped graph-based methods across five core stages of the cyber-attack detection pipeline—ranging from data acquisition to threat classification—offering a unified lens through which diverse research efforts can be compared. Despite notable progress, the review also uncovered significant challenges. These include limited standardization in datasets, a lack of real-time and scalable graph models, and the need for greater explainability and robustness in dynamic threat environments. Furthermore, encrypted traffic analysis and APT detection remain underexplored and demand more targeted attention. In conclusion, this study serves as both a roadmap and a research foundation for the continued development of graph-based cyber-attack detection systems. Future work should aim to bridge the gap between experimental success and operational feasibility by designing interpretable and context-aware graph learning models capable of addressing the evolving landscape of cybersecurity threats.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

References

1. M. Rabzelj, C. Bohak, L. S. Juznic, A. Kos, and U. Sedlar, "Cyber-attack Graph Modeling for Visual Analytics," *IEEE Access* 11 (2023): 86910–86944, https://doi.org/10.1109/ACCESS.2023.3304640.

2. K. A. Scarfone and P. M. Mell, "Guide to Intrusion Detection and Prevention Systems (IDPS)," 2007, https://www.nist.gov/publications/guide-intrusion-detection-and-prevention-systems-idps.

3. B. Wang, Y. Zheng, W. Lou, and Y. T. Hou, "DDoS Attack Protection in the Era of Cloud Computing and Software-Defined Networking," *Computer Networks* 81 (2015): 308–319, https://doi.org/10.1016/J.COMNET.2015.02.026.

4. P. Mishra, V. Varadharajan, U. Tupakula, and E. S. Pilli, "A Detailed Investigation and Analysis of Using Machine Learning Techniques for Intrusion Detection," *IEEE Communications Surveys and Tutorials* 21, no. 1 (2019): 686–728, https://doi.org/10.1109/COMST.2018.2847722.

5. M. Alkasassbeh and M. Almseidin, "Machine Learning Methods for Network Intrusion Detection," 2018, http://arxiv.org/abs/1809.02610.

6. D. Gibert, C. Mateu, and J. Planes, "The Rise of Machine Learning for Detection and Classification of Malware: Research Developments, Trends and Challenges," *Journal of Network and Computer Applications* 153 (2020): 102526, https://doi.org/10.1016/J.JNCA.2019.102526.

7. A. Shalaginov, S. Banin, A. Dehghantanha, and K. Franke, "Machine Learning Aided Static Malware Analysis: A Survey and Tutorial," *Advances in Information Security* 70 (2018): 7–45, https://doi.org/10.1007/978-3-319-73951-9_2.

8. H. Liu and B. Lang, "Machine Learning and Deep Learning Methods for Intrusion Detection Systems: A Survey," *Applied Sciences* 9, no. 20 (2019): 4396, https://doi.org/10.3390/APP9204396.

9. İ. Türker and A. M. Kanca, "A Bibliometric Comparison On Machine Learning vs. Deep Learning Approaches In Cybersecurity," https://www.researchgate.net/publication/366840630.

15320634, 2025, 27-28, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cpe.70389 by The Purchase Officer Purchase Office, Wiley Online Library on [01/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10. A. Grover and J. Leskovec, "Node2vec: Scalable Feature Learning for Networks," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, vol. 13–17 (Association for Computing Machinery (ACM), 2016), 855–864, https://doi.org/10.1145/2939672.2939754.

11. N. Shervashidze, S. V. N. Vishwanathan, T. H. Petri, K. Mehlhorn, and K. M. Borgwardt, "Efficient Graphlet Kernels for Large Graph Comparison," 2009.

12. R. Albert and A.-L. Barabasi, "Statistical Mechanics of Complex Networks," *Reviews of Modern Physics* 74, no. 1 (2001): 47–97, https://doi.org/10.1103/RevModPhys.74.47.

13. A. S. da Mata, "Complex Networks: a Mini-Review," *Brazilian Journal of Physics* 50, no. 5 (2020): 658–672, https://doi.org/10.1007/S13538-020-00772-9/FIGURES/15.

14. G. Zhong, L. N. Wang, X. Ling, and J. Dong, "An Overview on Data Representation Learning: From Traditional Feature Learning to Recent Deep Learning," *Journal of Finance and Data Science* 2, no. 4 (2016): 265–278, https://doi.org/10.1016/j.jfds.2017.05.001.

15. I. Turker and S. O. Tan, "Scientific Impact of Graph-Based Approaches in Deep Learning Studies—A Bibliometric Comparison," 2022, https://arxiv.org/pdf/2210.07343.

16. M. Xu, "Understanding Graph Embedding Methods and Their Applications," *SIAM Review* 63, no. 4 (2020): 825–853, https://doi.org/10.1137/20M1386062.

17. T. Danışmanı Doç İlker, "Türker, "Kelime Gömme Vektörlerinin Graf Dönüşümü Yoluyla Metin Siniflandirmada Kullanimi 2023 Yüksek Lisans Tezi Bilgisayar Mühendisliği Elif Dorukbaşi."

18. A. D. Pazho, G. A. Noghre, A. A. Purkayastha, J. Vempati, O. Martin, and H. Tabkhi, "A Survey of Graph-Based Deep Learning for Anomaly Detection in Distributed Systems," *IEEE Transactions on Knowledge and Data Engineering* 36, no. 1 (2024): 1–20, https://doi.org/10.1109/TKDE.2023.3282898.

19. S. Noel, "A Review of Graph Approaches to Network Security Analytics," in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 11170 LNCS* (Springer, 2018), 300–323, https://doi.org/10.1007/978-3-030-04834-1_16.

20. S. Noel, E. Harley, K. H. Tam, M. Limiero, and M. Share, "CyGraph: Graph-Based Analytics and Visualization for Cybersecurity," in *Handbook of Statistics*, vol. 35 (Elsevier B.V., 2016), 117–167, https://doi.org/10.1016/BS.HOST.2016.07.001.

21. Y. C. A. P. Reddy, P. Viswanath, and B. E. Reddy, "Semi-Supervised Learning: a Brief Review," *International Journal of Engineering & Technology* 7, no. 1.8 (2018): 81, https://doi.org/10.14419/ijet.v7i1.8.9977.

22. A. Géron, *Hands-On Machine Learning With Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems* (O'Reilly Media, Inc., 2019).

23. B. Alwasel, A. Aldribi, M. Alreshoodi, I. S. Alsukayti, and M. Alsuhaibani, "Leveraging Graph-Based Representations to Enhance Machine Learning Performance in IIoT Network Security and Attack Detection," *Applied Sciences* 13, no. 13 (2023): 7774, https://doi.org/10.3390/app13137774.

24. M. Baykara and R. Das, "A Novel Hybrid Approach for Detection of Web-Based Attacks in Intrusion Detection Systems," *International Journal of Computer Networks and Applications* 4, no. 2 (2017): 62, https://doi.org/10.22247/IJCNA/2017/48968.

25. H. Ahmetoglu and R. Das, "A Comprehensive Review on Detection of Cyber-Attacks: Data Sets, Methods, Challenges, and Future Research Directions," *Internet of Things* 20 (2022): 100615, https://doi.org/10.1016/J.IOT.2022.100615.

26. M. Shafiq, Z. Tian, Y. Sun, X. Du, and M. Guizani, "Selection of Effective Machine Learning Algorithm and Bot-IoT Attacks Traffic Identification for Internet of Things in Smart City," *Future Generation Computer Systems* 107 (2020): 433–442, https://doi.org/10.1016/J.FUTURE.2020.02.017.

27. M. Shafiq, Z. Tian, A. K. Bashir, X. Du, and M. Guizani, "CorrAUC: A Malicious Bot-IoT Traffic Detection Method in IoT Network Using Machine-Learning Techniques," *IEEE Internet of Things Journal* 8, no. 5 (2021): 3242–3254, https://doi.org/10.1109/JIOT.2020.3002255.

28. S. Pokhrel, R. Abbas, and B. Aryal, "IoT Security: Botnet Detection in IoT Using Machine Learning," 2021, https://arxiv.org/pdf/2104.02231.

29. R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, and S. Venkatraman, "Robust Intelligent Malware Detection Using Deep Learning," *IEEE Access* 7 (2019): 46717–46738, https://doi.org/10.1109/ACCESS.2019.2906934.

30. S. I. Popoola, B. Adebisi, M. Hammoudeh, G. Gui, and H. Gacanin, "Hybrid Deep Learning for Botnet Attack Detection in the Internet-Of-Things Networks," *IEEE Internet of Things Journal* 8, no. 6 (2021): 4944–4956, https://doi.org/10.1109/JIOT.2020.3034156.

31. D. Vasan, M. Alazab, S. Wassan, H. Naeem, B. Safaei, and Q. Zheng, "IMCFN: Image-Based Malware Classification Using Fine-Tuned Convolutional Neural Network Architecture," *Computer Networks* 171 (2020): 107138, https://doi.org/10.1016/J.COMNET.2020.107138.

32. J. Meira, R. Andrade, I. Praça, et al., "Performance Evaluation of Unsupervised Techniques in Cyber-Attack Anomaly Detection," *Journal of Ambient Intelligence and Humanized Computing* 11, no. 11 (2020): 4477–4489, https://doi.org/10.1007/S12652-019-01417-9.

33. Z. Alothman, M. Alkasassbeh, and S. Al-Haj Baddar, "An Efficient Approach to Detect IoT Botnet Attacks Using Machine Learning," *Journal of High Speed Networks* 26, no. 3 (2020): 241–254, https://doi.org/10.3233/JHS-200641.

34. G. Thamilarasu and S. Chawla, "Towards Deep-Learning-Driven Intrusion Detection for the Internet of Things," *Sensors* 19, no. 9 (2019): 1977, https://doi.org/10.3390/S19091977.

35. M. Kravchik and A. Shabtai, "Detecting Cyber Attacks in Industrial Control Systems Using Convolutional Neural Networks," in *Proceedings of the ACM Conference on Computer and Communications Security* (ACM, 2018), 72–83, https://doi.org/10.1145/3264888.3264896.

36. H. Zhang, L. Huang, C. Q. Wu, and Z. Li, "An Effective Convolutional Neural Network Based on SMOTE and Gaussian Mixture Model for Intrusion Detection in Imbalanced Dataset," *Computer Networks* 177 (2020): 107315, https://doi.org/10.1016/J.COMNET.2020.107315.

37. M. Asadi, "Detecting IoT Botnets Based on the Combination of Cooperative Game Theory With Deep and Machine Learning Approaches," *Journal of Ambient Intelligence and Humanized Computing* 13, no. 12 (2022): 5547–5561, https://doi.org/10.1007/S12652-021-03185-X/METRICS.

38. I. Apostol, M. Preda, C. Nila, and I. Bica, "IoT Botnet Anomaly Detection Using Unsupervised Deep Learning," *Electronics* 10, no. 16 (2021): 1876, https://doi.org/10.3390/ELECTRONICS10161876.

39. I. Ullah and Q. H. Mahmoud, "Design and Development of a Deep Learning-Based Model for Anomaly Detection in IoT Networks," *IEEE Access* 9 (2021): 103906–103926, https://doi.org/10.1109/ACCESS.2021.3094024.

40. S. Lagraa, M. Husák, H. Seba, S. Vuppala, R. State, and M. Ouedraogo, "A Review on Graph-Based Approaches for Network Security Monitoring and Botnet Detection," *International Journal of Information Security* 23, no. 1 (2024): 119–140, https://doi.org/10.1007/S10207-023-00742-7/METRICS.

41. L. Chang and P. Branco, "Graph-Based Solutions With Residuals for Intrusion Detection: the Modified E-GraphSAGE and E-ResGAT Algorithms," 2021, https://arxiv.org/pdf/2111.13597.

15320634, 2025, 27-28, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cpe.70389 by The Purchase Officer Purchase Office, Wiley Online Library on [01/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

42. F. Scarselli, M. Hagenbuchner, S. L. Yong, A. C. Tsoi, M. Gori, and M. Maggini, "Graph Neural Networks for Ranking Web Pages," in *Proceedings—2005 IEEE/WIC/ACM InternationalConference on Web Intelligence, WI 2005* (IEEE (Institute of Electrical and Electronics Engineers), 2005), 666–672, https://doi.org/10.1109/WI.2005.67.

43. P. Yanardag and S. V. N. Vishwanathan, "Deep graph kernels," *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 2015 (2015): 1365–1374, https://doi.org/10.1145/2783258.2783417/SUPPL_FILE/P1365.MP4.

44. N. Ghadiri, R. Samani, and F. Shahrokh, "Integration of Text and Graph-Based Features for Depression Detection Using Visibility Graph," in *Lecture Notes in Networks and Systems, 646 LNNS* (Springer Nature Switzerland AG, 2023), 332–341, https://doi.org/10.1007/978-3-031-27440-4_32.

45. H. İ. Aslan and C. Choi, "VisGIN: Visibility Graph Neural Network on One-Dimensional Data for Biometric Authentication," *Expert Systems with Applications* 237 (2024): 121323, https://doi.org/10.1016/J.ESWA.2023.121323.

46. L. Lacasa, B. Luque, F. Ballesteros, J. Luque, and J. C. Nuno, "From Time Series to Complex Networks: the Visibility Graph," *Proceedings of the National Academy of Sciences of the United States of America* 105, no. 13 (2008): 4972–4975, https://doi.org/10.1073/pnas.0709247105.

47. C. Bergillos Varela and A. Arratia Quesada, "A Study of Visibility Graphs for Time Series Representations," 2020.

48. T. Wen, H. Chen, and K. H. Cheong, "Visibility Graph for Time Series Prediction and Image Classification: a Review," *Nonlinear Dynamics* 110, no. 4 (2022): 2979–2999, https://doi.org/10.1007/S11071-022-08002-4/METRICS.

49. B. Luque, L. Lacasa, F. Ballesteros, and J. Luque, "Horizontal Visibility Graphs: Exact Results for Random Time Series," *Physical Review. E, Statistical, Nonlinear, and Soft Matter Physics* 80, no. 4 (2010): 046103, https://doi.org/10.1103/PhysRevE.80.046103.

50. S. Gamage and J. Samarabandu, "Deep Learning Methods in Network Intrusion Detection: A Survey and an Objective Comparison," *Journal of Network and Computer Applications* 169 (2020): 102767, https://doi.org/10.1016/J.JNCA.2020.102767.

51. A. A. Diro and N. Chilamkurti, "Distributed Attack Detection Scheme Using Deep Learning Approach for Internet of Things," *Future Generation Computer Systems* 82 (2018): 761–768, https://doi.org/10.1016/J.FUTURE.2017.08.043.

52. D. Zhao, I. Traore, B. Sayed, et al., "Botnet Detection Based on Traffic Behavior Analysis and Flow Intervals," *Computers & Security* 39, no. PARTA (2013): 2–16, https://doi.org/10.1016/J.COSE.2013.04.007.

53. A. Aldweesh, A. Derhab, and A. Z. Emam, "Deep Learning Approaches for Anomaly-Based Intrusion Detection Systems: A Survey, Taxonomy, and Open Issues," *Knowledge-Based Systems* 189 (2020): 105124, https://doi.org/10.1016/J.KNOSYS.2019.105124.

54. T. Gangavarapu, C. D. Jaidhar, and B. Chanduka, "Applicability of Machine Learning in Spam and Phishing Email Filtering: Review and Approaches," *Artificial Intelligence Review* 53 (2020): 5019–5081, https://doi.org/10.1007/s10462-020-09814-9.

55. P. Y. Chen, S. Choudhury, and A. O. Hero, "Multi-Centrality Graph Spectral Decompositions and Their Application to Cyber Intrusion Detection," *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing—Proceedings* 2016 (2016): 4553–4557, https://doi.org/10.1109/ICASSP.2016.7472539.

56. L. C. Freeman, "A Set of Measures of Centrality Based on Betweenness," *Sociometry* 40, no. 1 (1977): 35, https://doi.org/10.2307/3033543.

57. G. Sabidussi, "The Centrality Index of a Graph," *Psychometrika* 31, no. 4 (1966): 581–603, https://doi.org/10.1007/BF02289527/METRICS.

58. A. Bihari and M. K. Pandia, "Eigenvector Centrality and Its Application in Research Professionals' Relationship Network," *2015 1st International Conference on Futuristic Trends in Computational Analysis and Knowledge Management, ABLAZE 2015* (IEEE (Institute of Electrical and Electronics Engineers), 2015), 510–514, https://doi.org/10.1109/ABLAZE.2015.7154915.

59. X. Bi, G. Chen, and G. Lu, "Visibility Graph Based Feature Extraction for Fault Diagnosis of Rolling Bearings," *Global Reliability and Prognostics and Health Management, PHM-Nanjing 2021, 2021* (IEEE (Institute of Electrical and Electronics Engineers), 2021), https://doi.org/10.1109/PHM-NANJING52125.2021.9613128.

60. P. Dharani Devi and R. Thanuja, "Convolutional Neural Network Based Deep Feature Extraction in Remote Sensing Images," *Proceedings—International Conference on Smart Electronics and Communication, ICOSEC* 2020 (2020): 441–444, https://doi.org/10.1109/ICOSEC49089.2020.9215274.

61. J. Zhang, J. Xia, X. Liu, and J. Olichney, "Machine Learning on Visibility Graph Features Discriminates the Cognitive Event-Related Potentials of Patients With Early Alzheimer's Disease From Healthy Aging," *Brain Sciences* 13, no. 5 (2023): 770, https://doi.org/10.3390/BRAINSCI13050770.

62. İ. Türker, "Sars Cov-2 CT-Scan Image Classification Using Graph Convolutional Networks," www.istanbulkongresi.org.

63. A. Piórkowski, R. Obuchowicz, K. Nurzynska, P. Q. Rashid, and I. Türker, "Lung Disease Detection Using U-Net Feature Extractor Cascaded by Graph Convolutional Network," *Diagnostics* 14, no. 12 (2024): 1313, https://doi.org/10.3390/DIAGNOSTICS14121313.

64. D. F. P. Melo, I. S. Fadigas, and H. B. B. Pereira, "Graph-Based Feature Extraction: A New Proposal to Study the Classification of Music Signals Outside the Time-Frequency Domain," *PLoS One* 15, no. 11 (2020): e0240915, https://doi.org/10.1371/JOURNAL.PONE.0240915.

65. B. Yan, C. Yang, C. Shi, et al., "Graph Mining for Cybersecurity: A Survey," *ACM Transactions on Knowledge Discovery from Data* 18 (2023): 1–52, https://doi.org/10.1145/3610228.

66. W. Jiang, "Graph-Based Deep Learning for Communication Networks: A Survey," *Computer Communications* 185 (2022): 40–54, https://doi.org/10.1016/j.comcom.2021.12.015.

67. T. Pourhabibi, K. L. Ong, B. H. Kam, and Y. L. Boo, "Fraud Detection: A Systematic Literature Review of Graph-Based Anomaly Detection Approaches," *Decision Support Systems* 133 (2020): 113303, https://doi.org/10.1016/j.dss.2020.113303.

68. Y. Lv, S. Qin, Z. Zhu, Z. Yu, S. Li, and W. Han, "A Review of Provenance Graph Based APT Attack Detection:Applications and Developments," in *Proceedings—2022 7th IEEE International Conference on Data Science in Cyberspace, DSC 2022* (Institute of Electrical and Electronics Engineers Inc, 2022), 498–505, https://doi.org/10.1109/DSC55868.2022.00075.

69. T. Bilot, N. El Madhoun, K. Al Agha, and A. Zouaoui, "Graph Neural Networks for Intrusion Detection: A Survey," *IEEE Access* 11 (2023): 49114–49139, https://doi.org/10.1109/ACCESS.2023.3275789.

70. H. Zhu and J. Lu, "Graph-Based Intrusion Detection System Using General Behavior Learning," in *2022 IEEE Global Communications Conference, GLOBECOM 2022—Proceedings* (Institute of Electrical and Electronics Engineers Inc, 2022), 2621–2626, https://doi.org/10.1109/GLOBECOM48099.2022.10001299.

71. Y. Yao, L. Su, Z. Lu, and B. Liu, "STDeepGraph: Spatial-Temporal Deep Learning on Communication Graphs for Long-Term Network Attack Detection," in *Proceedings—2019 18th IEEE International Conference on Trust, Security and Privacy in Computing and Communications/13th IEEE International Conference on Big Data Science and Engineering, TrustCom/BigDataSE 2019* (Institute of Electrical and Electronics Engineers Inc, 2019), 120–127, https://doi.org/10.1109/TrustCom/BigDataSE.2019.00025.

15320634, 2025, 27-28, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cpe.70389 by The Purchase Officer Purchase Office, Wiley Online Library on [01/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

72. L. Su, Y. Yao, Z. Lu, and B. Liu, "Understanding the Influence of Graph Kernels on Deep Learning Architecture: A Case Study of Flow-Based Network Attack Detection," in *Proceedings—2019 18th IEEE International Conference on Trust, Security and Privacy in Computing and Communications/13th IEEE International Conference on Big Data Science and Engineering, TrustCom/BigDataSE 2019* (Institute of Electrical and Electronics Engineers Inc, 2019), 312–318, https://doi.org/10.1109/TrustCom/BigDataSE.2019.00049.

73. L. Leichtnam, E. Totel, N. Prigent, and L. Mé, "Sec2graph: Network Attack Detection Based on Novelty Detection on Graph Structured Data," in *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* (Springer, 2020), 238–258, https://doi.org/10.1007/978-3-030-52683-2_12.

74. F. Zola, L. Segurola-Gil, J. L. Bruse, M. Galar, and R. Orduna-Urrutia, "Network Traffic Analysis Through Node Behaviour Classification: a Graph-Based Approach With Temporal Dissection and Data-Level Preprocessing," *Computers & Security* 115 (2022): 102632, https://doi.org/10.1016/j.cose.2022.102632.

75. Y. Yao, L. Su, and Z. Lu, "DeepGFL: Deep Feature Learning via Graph for Attack Detection on Flow-Based Network Traffic," in *Proceedings—IEEE Military Communications Conference MILCOM* (Institute of Electrical and Electronics Engineers Inc, 2018), 579–584, https://doi.org/10.1109/MILCOM.2018.8599821.

76. V. Desnitsky, D. Levshun, and I. Kotenko, "An Approach to Attack Detection on Graph-Based Modeling of States in Critical Infrastructures," in *Proceedings—2023 International Russian Automation Conference, RusAutoCon 2023* (Institute of Electrical and Electronics Engineers Inc, 2023), 420–424, https://doi.org/10.1109/RusAutoCon58002.2023.10272943.

77. E. Caville, W. W. Lo, S. Layeghy, and M. Portmann, "Anomal-E: A Self-Supervised Network Intrusion Detection System Based on Graph Neural Networks," *Knowledge-Based Systems* 258 (2022): 110030, https://doi.org/10.1016/j.knosys.2022.110030.

78. P. Sathishkumar, S. Nikitha, R. Sruthi, and R. R. Vishwa, "Anomaly Detection in Network Security Using Unsupervised Graph Neural Network," in *2025 International Conference on Advanced Computing Technologies (ICoACT)* (IEEE, 2025), 1–7, https://doi.org/10.1109/ICOACT63339.2025.11004728.

79. W. Wang, Y. Shang, Y. He, Y. Li, and J. Liu, "BotMark: Automated Botnet Detection With Hybrid Analysis of Flow-Based and Graph-Based Traffic Behaviors," *Information Sciences (N Y)* 511 (2020): 284–296, https://doi.org/10.1016/j.ins.2019.09.024.

80. K. Millar, L. Simpson, A. Cheng, H. G. Chew, and C. C. Lim, "Detecting Botnet Victims Through Graph-Based Machine Learning," in *Proceedings—International Conference on Machine Learning and Cybernetics* (IEEE Computer Society, 2021), https://doi.org/10.1109/ICMLC54886.2021.9737249.

81. K. Shinan, K. Alsubhi, and M. Usman Ashraf, "BotSward: Centrality Measures for Graph-Based Bot Detection Using Machine Learning," *Computers, Materials and Continua* 74, no. 1 (2023): 693–714, https://doi.org/10.32604/cmc.2023.031641.

82. K. Sinha, A. Viswanathan, and J. Bunn, "Tracking Temporal Evolution of Network Activity for Botnet Detection," 2019, https://arxiv.org/abs/1908.03443v1.

83. A. Alharbi and K. Alsubhi, "Botnet Detection Approach Using Graph-Based Machine Learning," *IEEE Access* 9 (2021): 99166–99180, https://doi.org/10.1109/ACCESS.2021.3094183.

84. A. A. Daya, M. A. Salahuddin, N. Limam, and R. Boutaba, "A Graph-Based Machine Learning Approach for Bot Detection," in *2019 IFIP/IEEE Symposium on Integrated Network and Service Management, IM 2019* (IEEE, 2019), 144–152, https://arxiv.org/abs/1902.08538v1.

85. S. Chowdhury, M. Khanzadeh, R. Akula, et al., "Botnet Detection Using Graph-Based Feature Clustering," *Journal of Big Data* 4, no. 1 (2017): 14, https://doi.org/10.1186/S40537-017-0074-7.

86. W. W. Lo, S. Layeghy, M. Sarhan, M. Gallagher, and M. Portmann, "E-GraphSAGE: A Graph Neural Network Based Intrusion Detection System for IoT," in *Proceedings of the IEEE/IFIP Network Operations and Management Symposium 2022: Network and Service Management in the era of Cloudification, Softwarization and Artificial Intelligence, NOMS 2022* (Institute of Electrical and Electronics Engineers Inc, 2022), https://doi.org/10.1109/NOMS54207.2022.9789878.

87. O. I. Khalaf, K. A. Ogudo, and S. K. B. Sangeetha, "Design of Graph-Based Layered Learning-Driven Model for Anomaly Detection in Distributed Cloud IoT Network," *Mobile Information Systems* 2022, no. 1 (2022): 6750757, https://doi.org/10.1155/2022/6750757.

88. C. Baru, "Institute of Electrical and Electronics Engineers, and IEEE Computer Society," in *2019 IEEE International Conference on Big Data: Proceedings* (IEEE, 2019).

89. X. Liu, J. Ren, H. He, B. Zhang, C. Song, and Y. Wang, "A Fast All-Packets-Based DDoS Attack Detection Approach Based on Network Graph and Graph Kernel," *Journal of Network and Computer Applications* 185 (2021): 103079, https://doi.org/10.1016/j.jnca.2021.103079.

90. A. Protogerou, S. Papadopoulos, A. Drosou, D. Tzovaras, and I. Refanidis, "A Graph Neural Network Method for Distributed Anomaly Detection in IoT," *Evolving Systems* 12, no. 1 (2021): 19–36, https://doi.org/10.1007/s12530-020-09347-0.

91. C. Wang and H. Zhu, "Wrongdoing Monitor: A Graph-Based Behavioral Anomaly Detection in Cyber Security," *IEEE Transactions on Information Forensics and Security* 17 (2022): 2703–2718, https://doi.org/10.1109/TIFS.2022.3191493.

92. H. Jing and J. Wang, "DDoS Detection Based on Graph Structure Features and Non-Negative Matrix Factorization," in *Concurrency and Computation: Practice and Experience*, vol. 34 (John Wiley and Sons Ltd, 2022), e5783, https://doi.org/10.1002/cpe.5783.

93. H. Jing and J. Wang, "Detection of DDoS Attack Within Industrial IoT Devices Based on Clustering and Graph Structure Features," *Security and Communication Networks* 2022 (2022): 1–9, https://doi.org/10.1155/2022/1401683.

94. W. Niu, R. Cao, X. Zhang, K. Ding, K. Zhang, and T. Li, "Opcode-Level Function Call Graph Based Android Malware Classification Using Deep Learning," *Sensors (Basel)* 20, no. 13 (2020): 1–23, https://doi.org/10.3390/s20133645.

95. Y. Wu, J. Shi, P. Wang, D. Zeng, and C. Sun, "DeepCatra: Learning Flow- and Graph-Based Behaviours for Android Malware Detection," *IET Information Security* 17, no. 1 (2023): 118–130, https://doi.org/10.1049/ise2.12082.

96. A. Cuzzocrea, M. Quebrado, A. Hafsaoui, and E. Serra, "A Graph-Representation-Learning Framework for Supporting Android Malware Identification and Polymorphic Evolution," in *Proceedings—2023 10th IEEE Swiss Conference on Data Science, SDS 2023* (Institute of Electrical and Electronics Engineers Inc, 2023), 34–41, https://doi.org/10.1109/SDS57534.2023.00012.

97. A. E. Karrar, "Adopting Graph-Based Machine Learning Algorithms to Classify Android Malware," *IJCSNS International Journal of Computer Science and Network Security* 22, no. 9 (2022): 840, https://doi.org/10.22937/IJCSNS.2022.22.9.109.

98. Y. Xin, L. Kong, Z. Liu, et al., "Machine Learning and Deep Learning Methods for Cybersecurity," *IEEE Access* 6 (2018): 35365–35381, https://doi.org/10.1109/ACCESS.2018.2836950.

99. Z. Li, X. Cheng, L. Sun, J. Zhang, and B. Chen, "A Hierarchical Approach for Advanced Persistent Threat Detection With Attention-Based Graph Neural Networks," *Security and Communication Networks* 2021 (2021): 1–14, https://doi.org/10.1155/2021/9961342.

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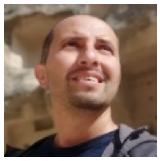
100. Z. Fu, M. Liu, Y. Qin, et al., "Encrypted Malware Traffic Detection via Graph-Based Network Analysis," in *ACM International Conference Proceeding Series* (Association for Computing Machinery, 2022), 495–509, https://doi.org/10.1145/3545948.3545983.

101. H. Liu and R. Jiang, "A Causal Graph-Based Approach for APT Predictive Analytics," *Electronics (Switzerland)* 12, no. 8 (2023): 1849, https://doi.org/10.3390/electronics12081849.

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